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Hsiao et al.

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(54) **METHOD OF MANUFACTURING A SEMICONDUCTOR DEVICE AND A SEMICONDUCTOR DEVICE**

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Related U.S. Application Data

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H01L 21/3105 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **H10D 64/01** (2025.01); **H01L 21/31051** (2013.01); **H01L 21/31116** (2013.01); **H01L 22/26** (2013.01); **H10D 64/017** (2025.01)

(58) **Field of Classification Search**

CPC H01L 29/401; H01L 21/31051; H01L 21/31116; H01L 22/26; H01L 29/66545
See application file for complete search history.

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Primary Examiner — Zandra V Smith

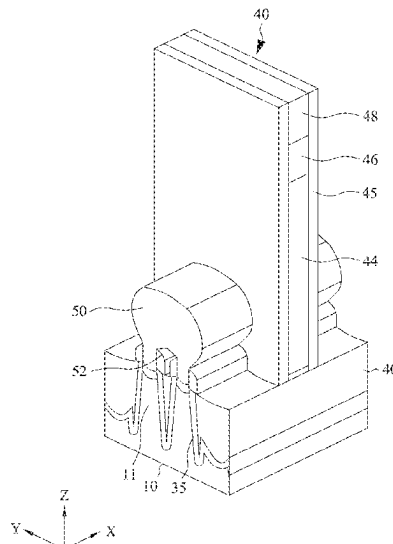
Assistant Examiner — Andre C Stevenson

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(57) **ABSTRACT**

In a method of manufacturing a semiconductor device, a sacrificial gate structure including sacrificial gate electrode is formed over a substrate. A first dielectric layer is formed over the sacrificial gate structure. A second dielectric layer is formed over the first dielectric layer. The second and first dielectric layers are planarized and recessed, and an upper portion of the sacrificial gate structure is exposed. A third dielectric layer is formed over the exposed sacrificial gate structure and over the first dielectric layer. A fourth dielectric layer is formed over the third dielectric layer. The fourth and third dielectric layers are planarized, and the sacrificial gate electrode is exposed and part of the third dielectric layer remains on the recessed first dielectric layer. The recessing the first dielectric layer comprises a first etching operation and a second etching operation using a different etching as from the first etching operation.

20 Claims, 25 Drawing Sheets



- (51) **Int. Cl.**
H01L 21/311 (2006.01)
H01L 21/66 (2006.01)

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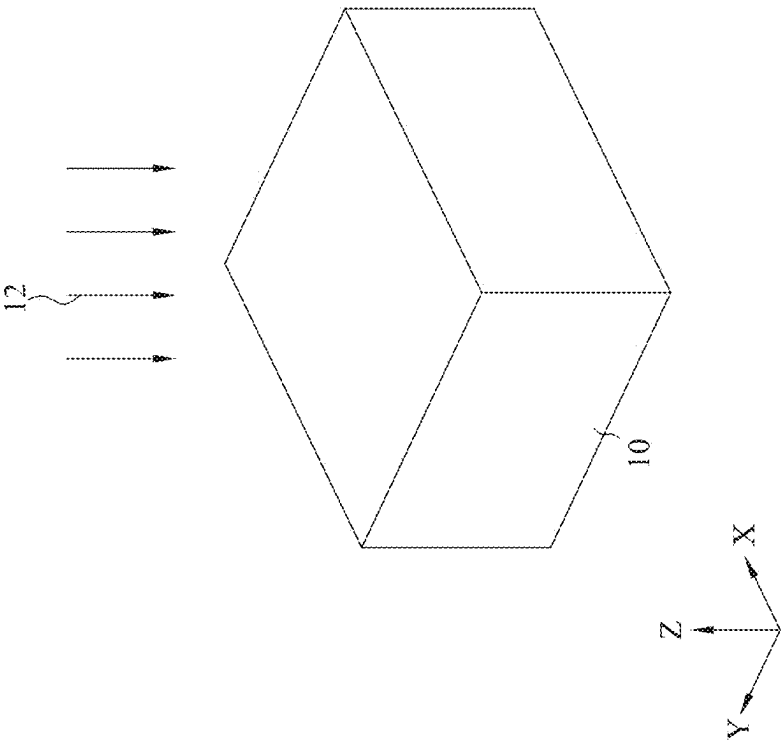


Fig. 1

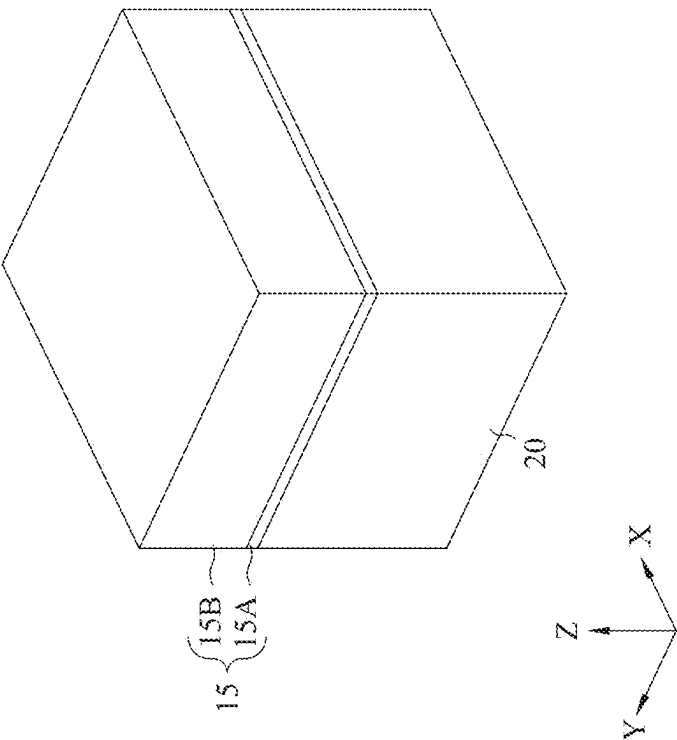


Fig. 2

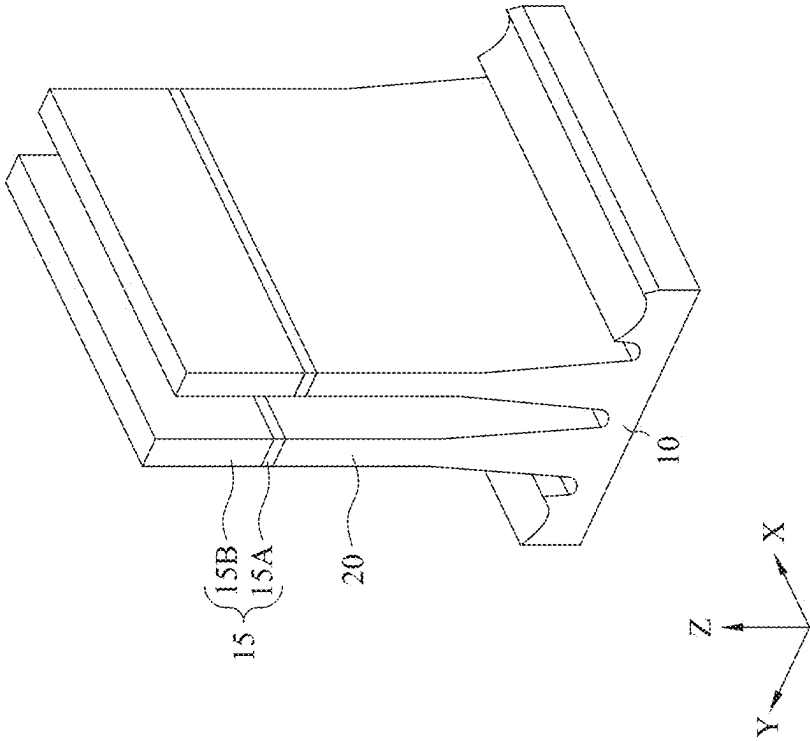


Fig. 3

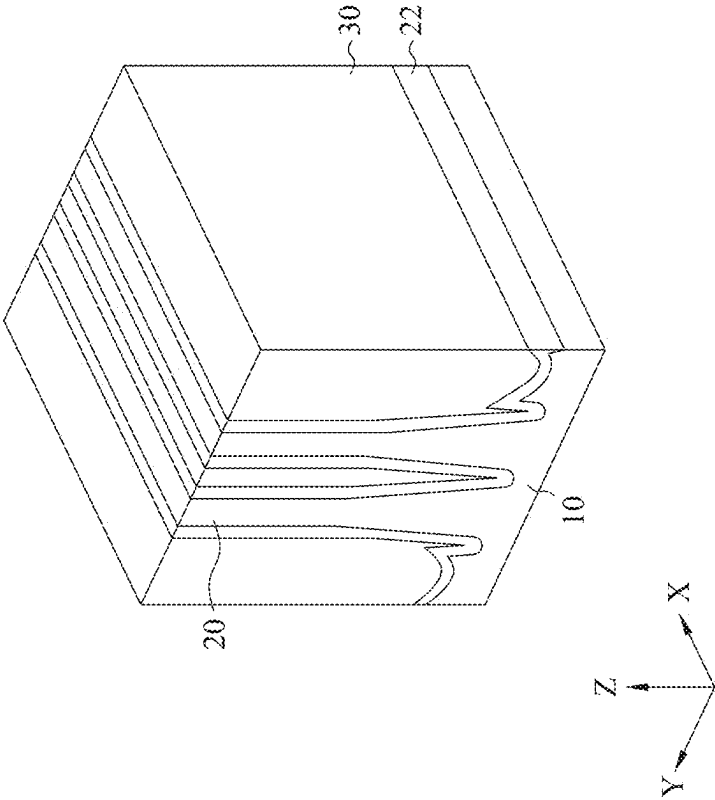


Fig. 4

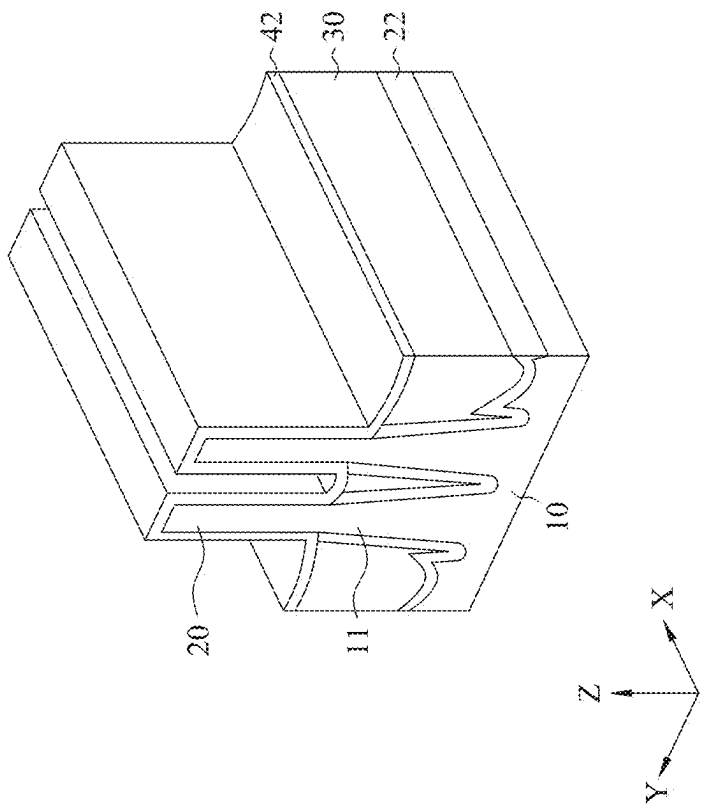


Fig. 5

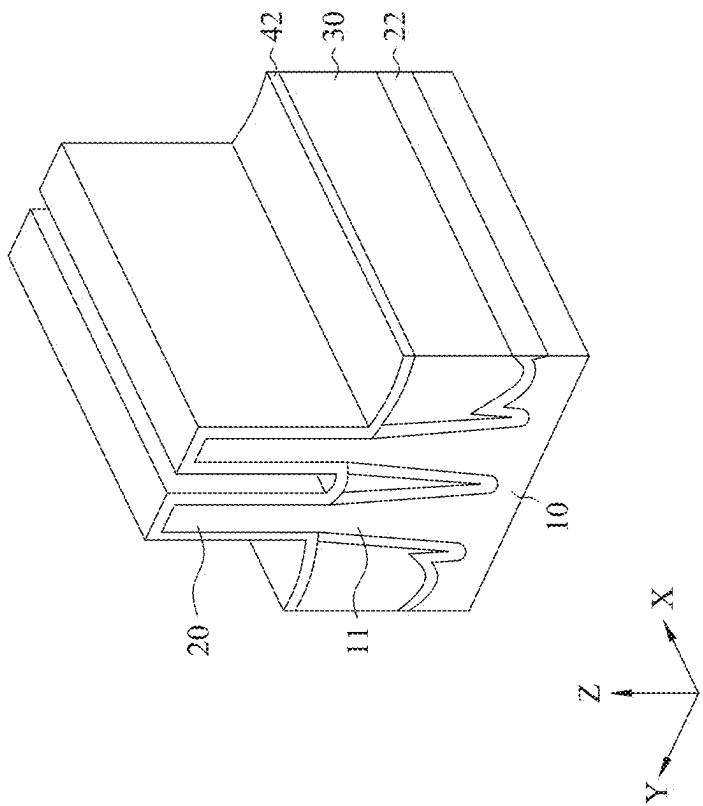


Fig. 6

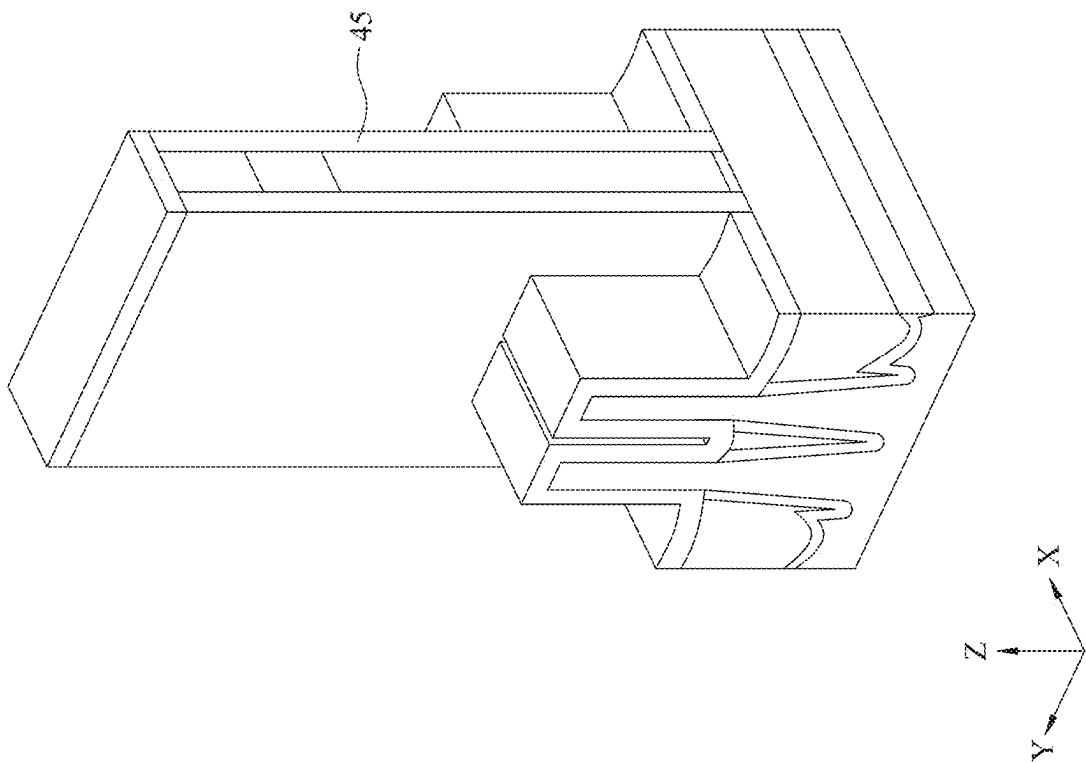


Fig. 8

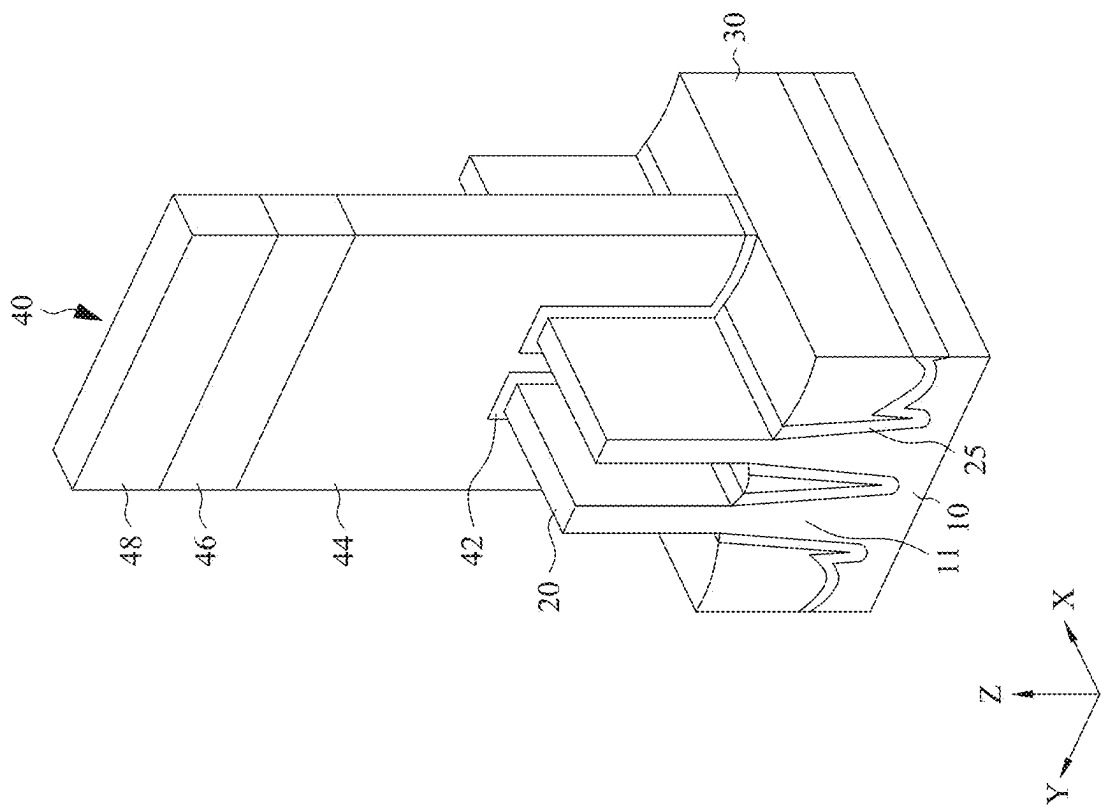


Fig. 7

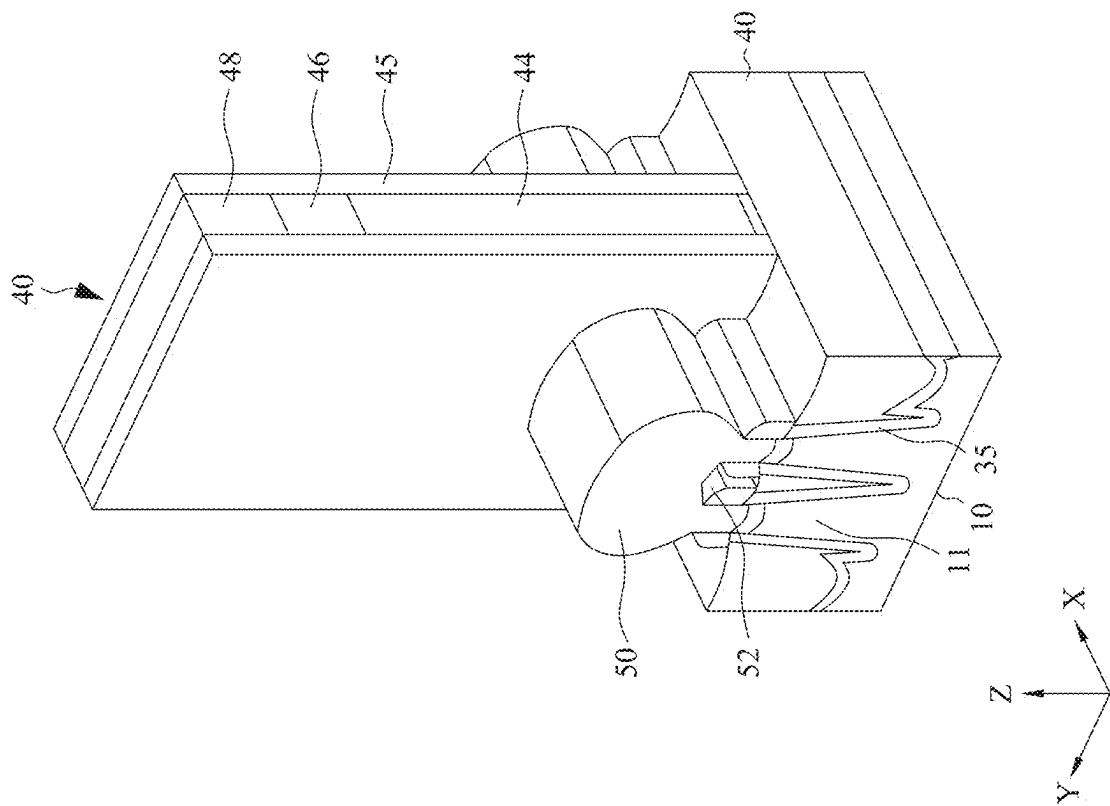


Fig. 9

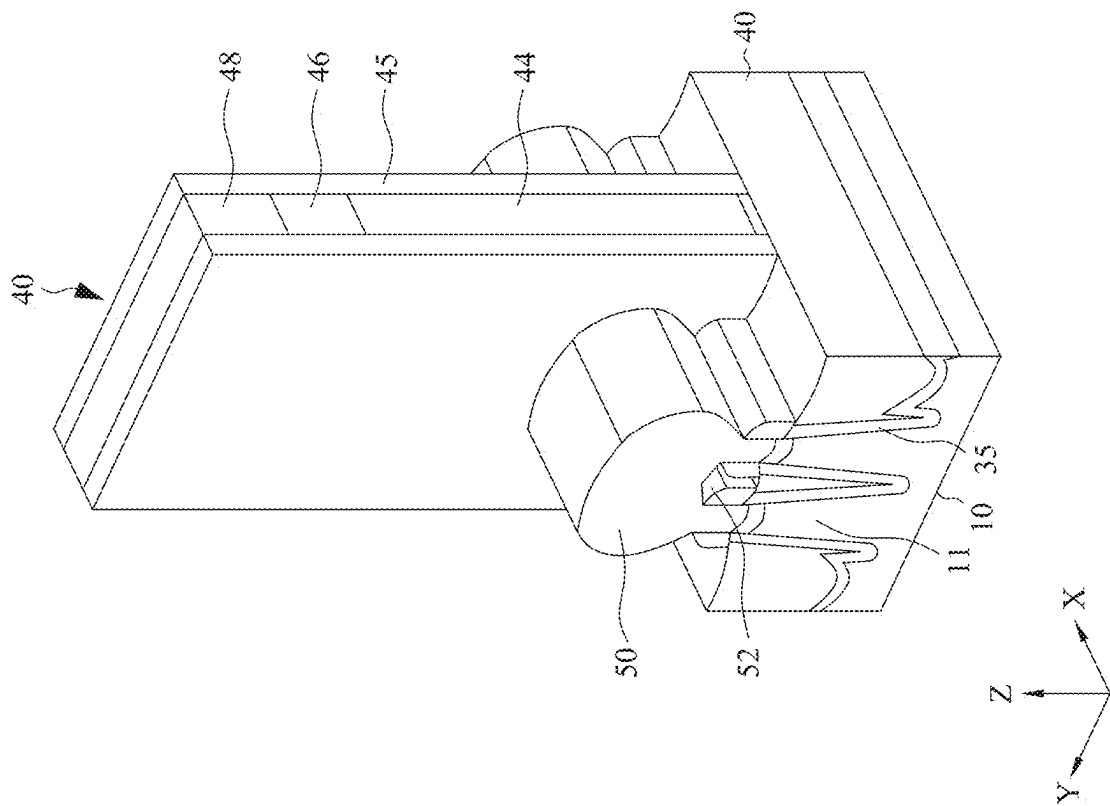


Fig. 10

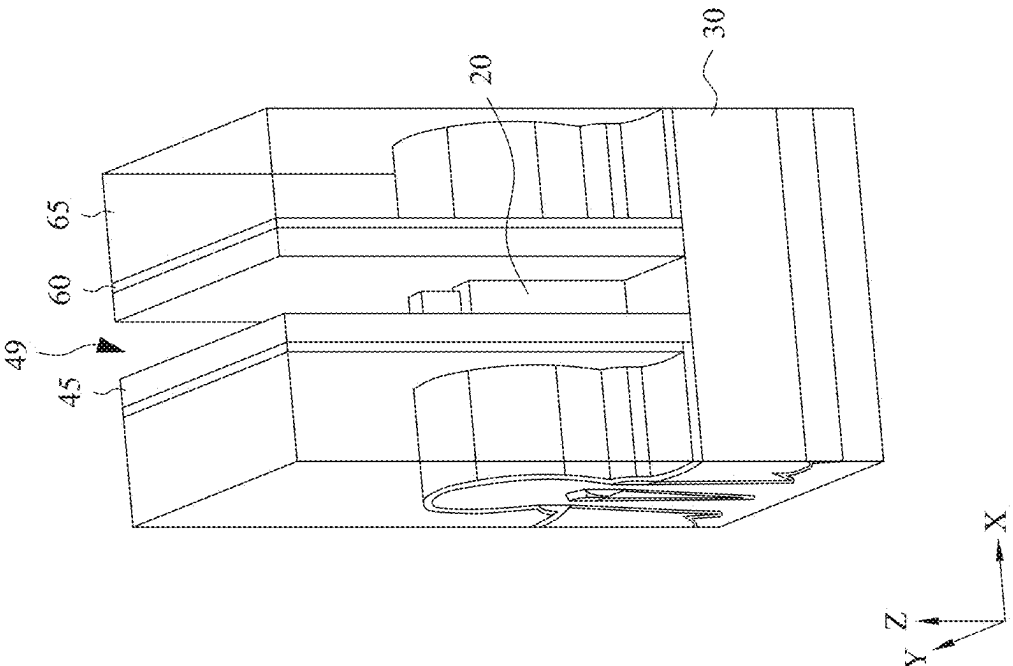


Fig. 11

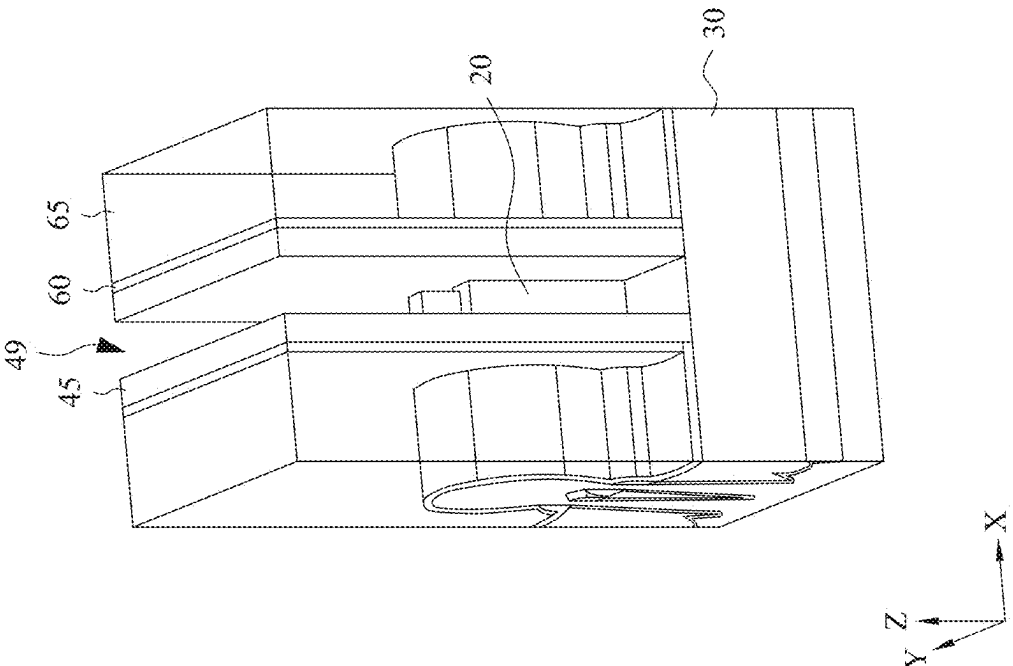


Fig. 12

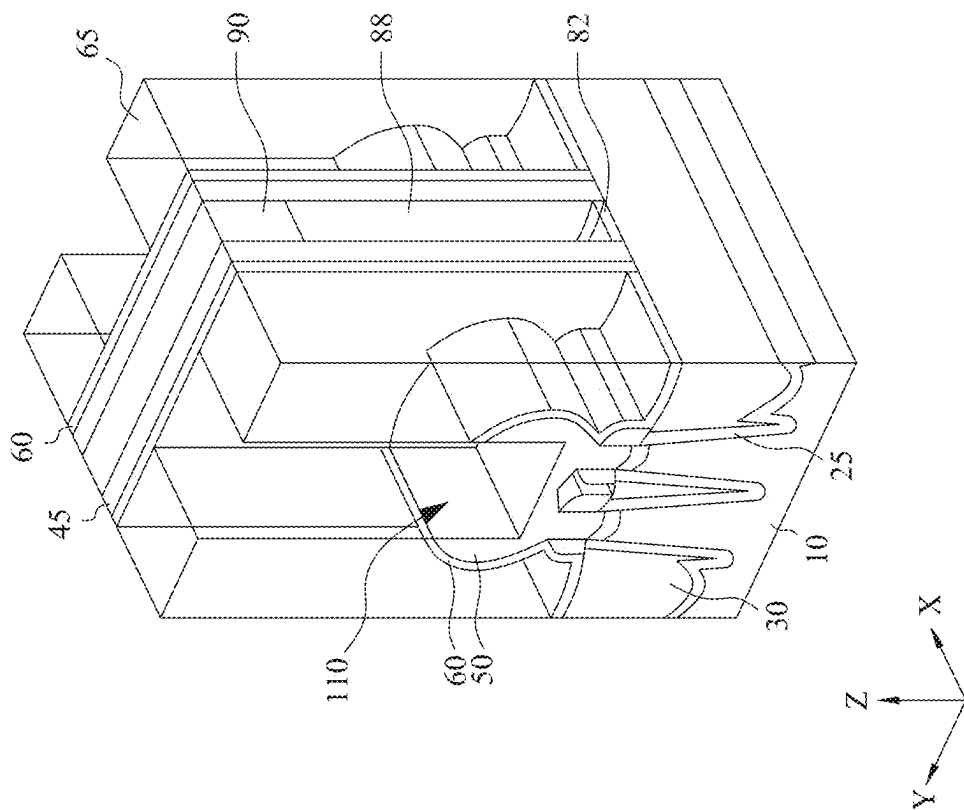


Fig. 13

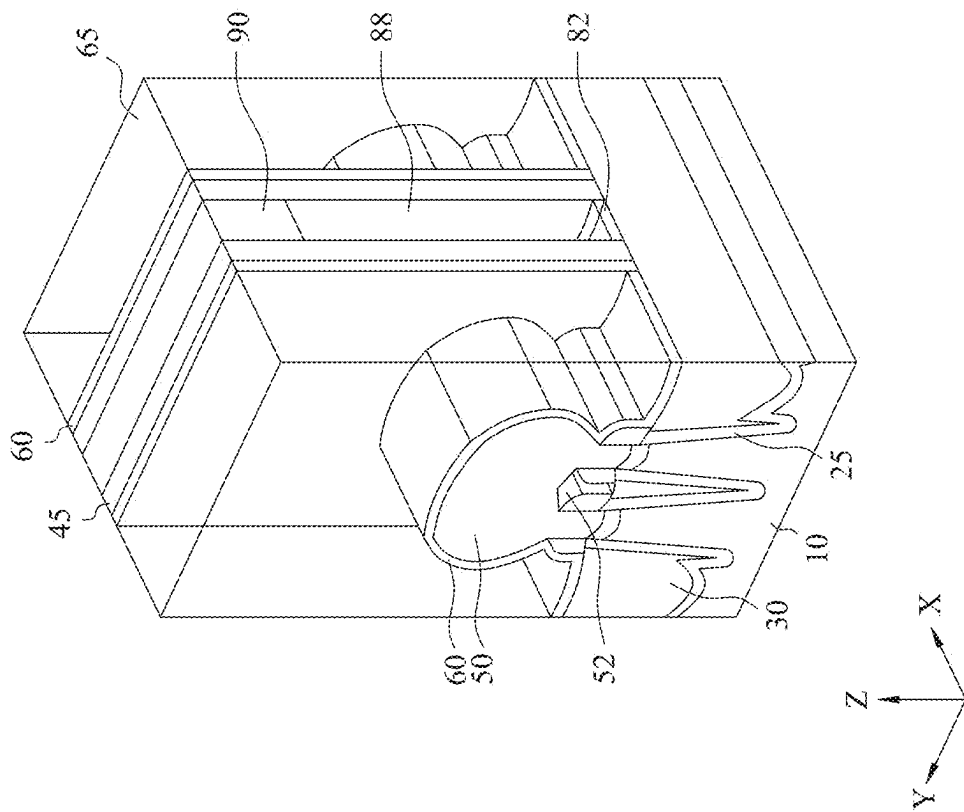
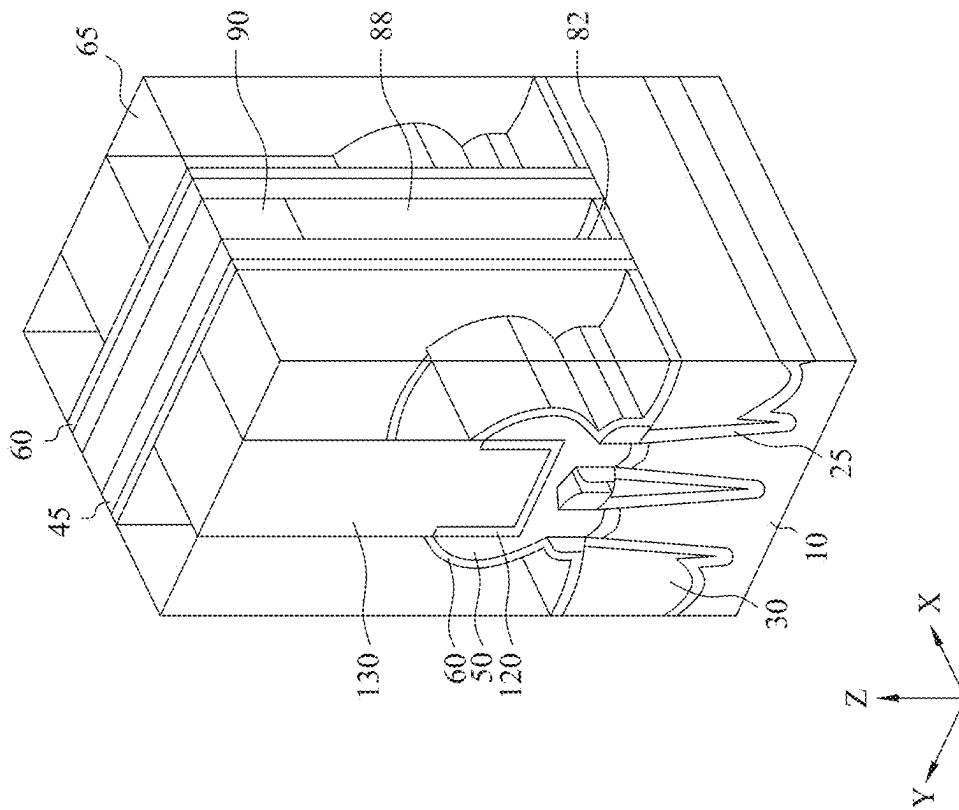
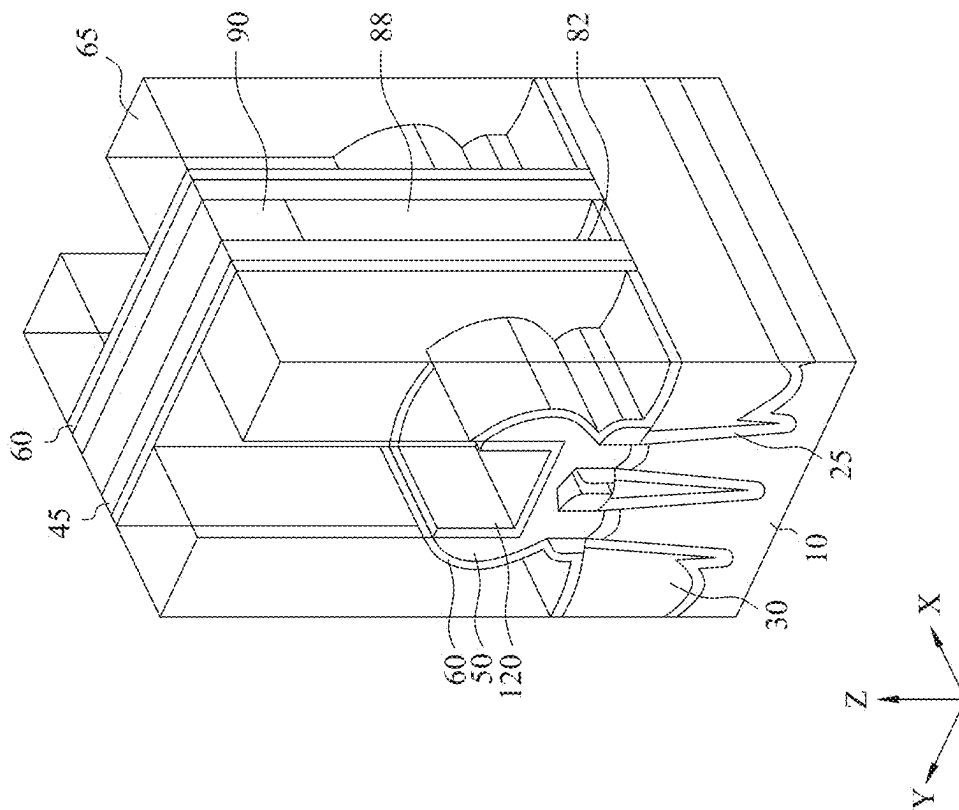
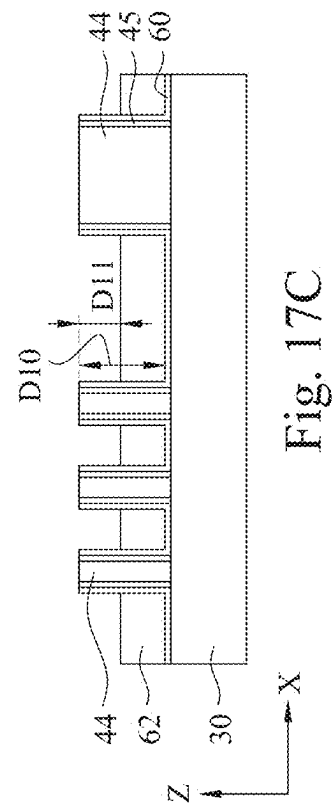
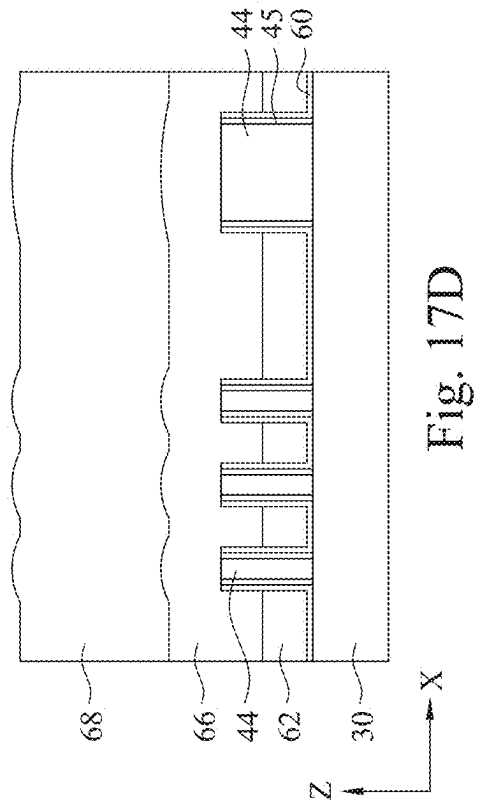
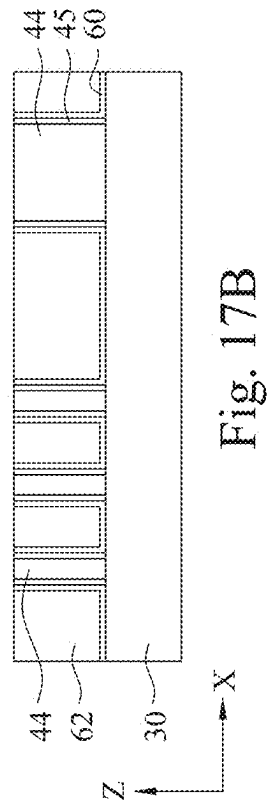
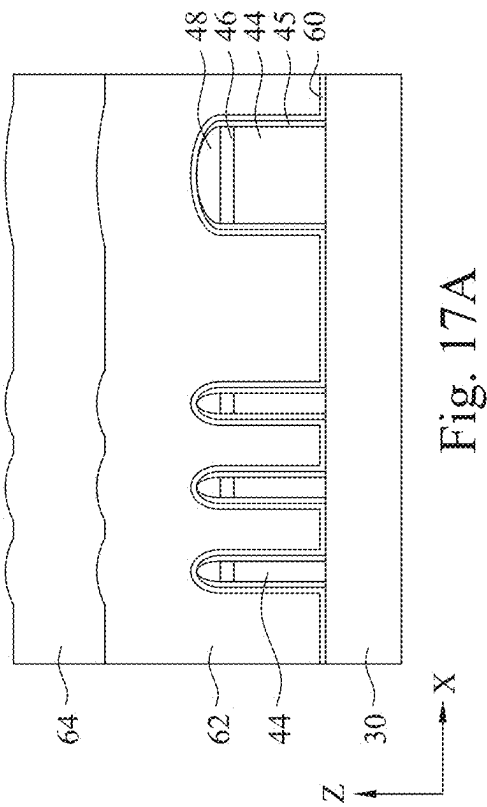


Fig. 14



60
11

50
 11



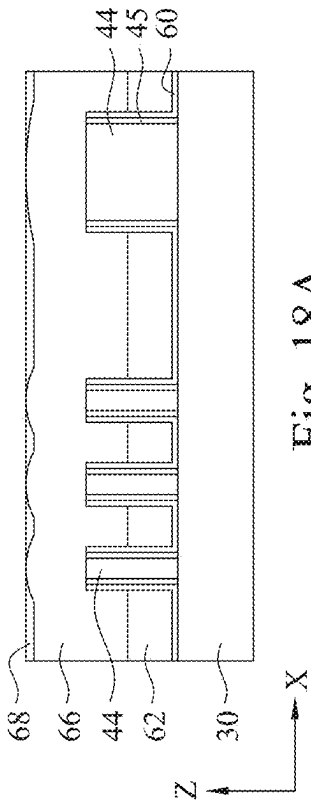


Fig. 18A

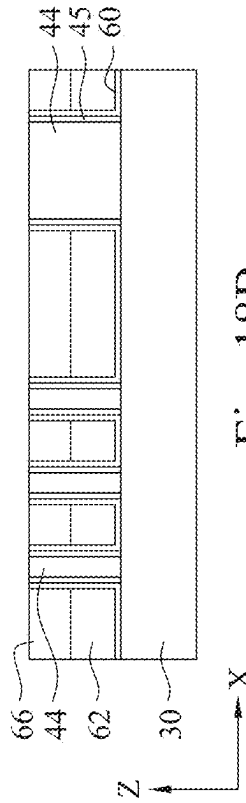


Fig. 18B

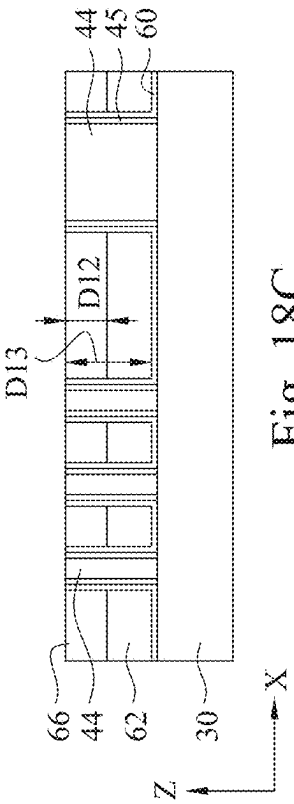


Fig. 18C

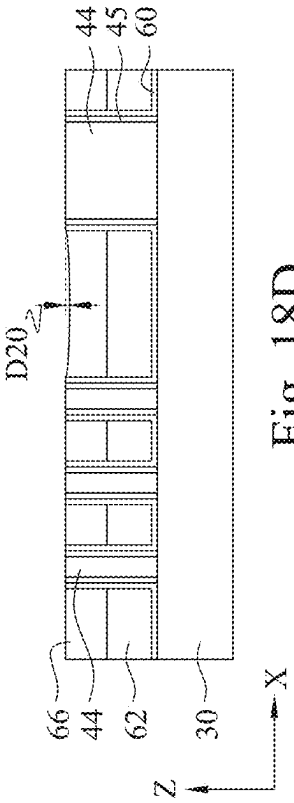


Fig. 18D

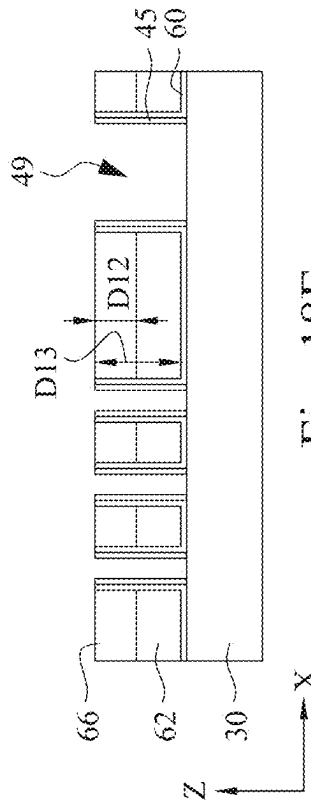


Fig. 18E

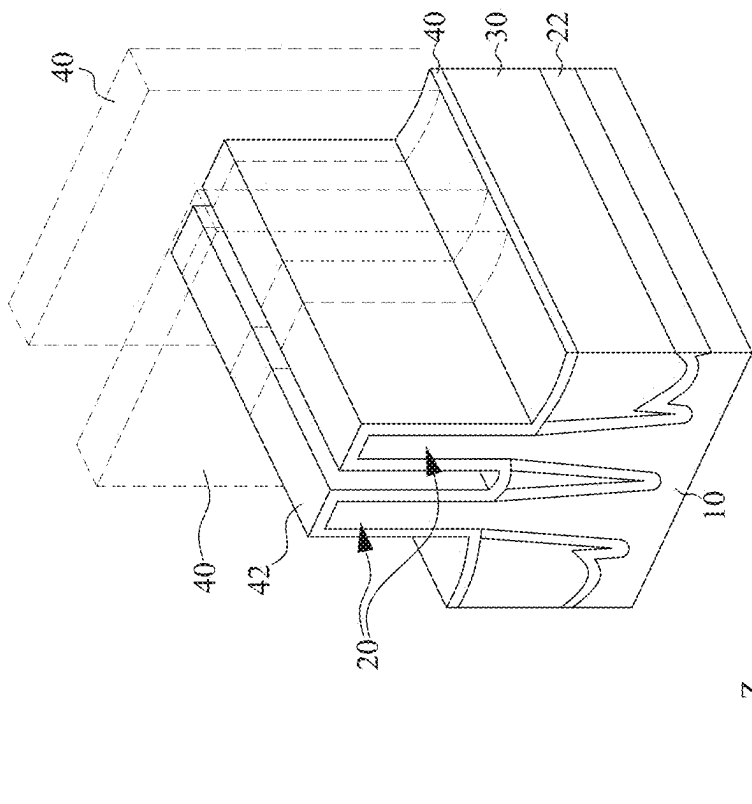


Fig. 19A

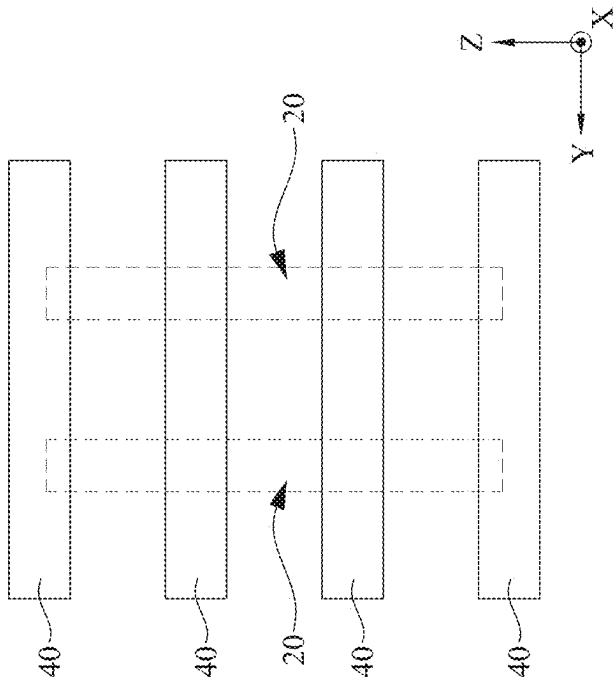
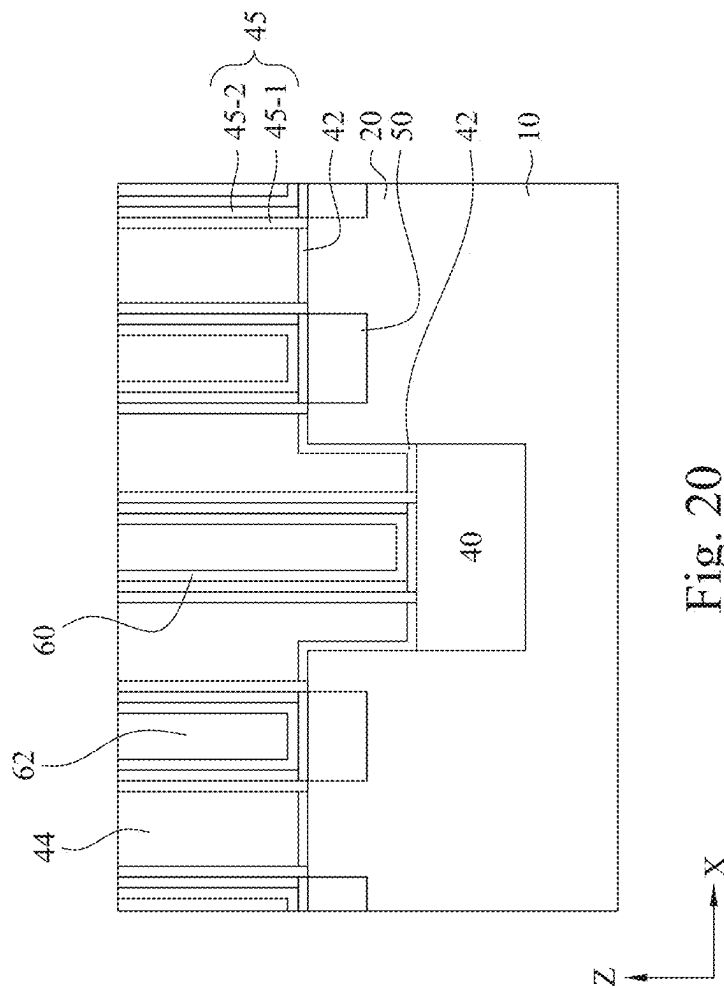


Fig. 19B



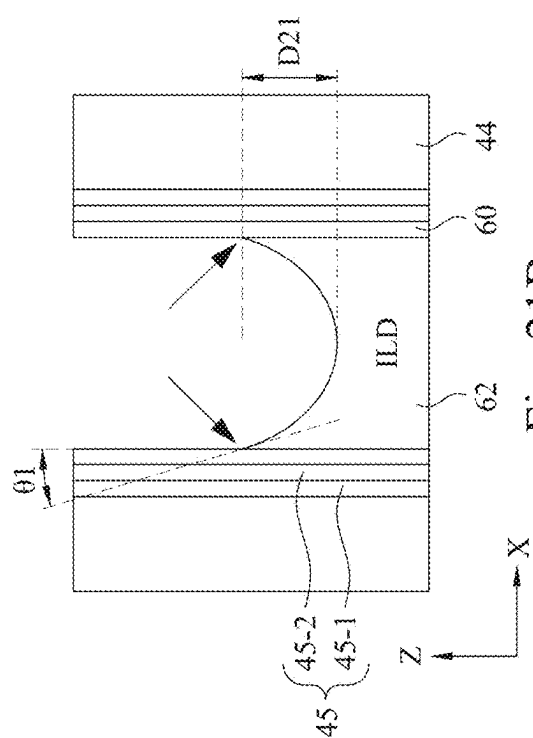


Fig. 21B

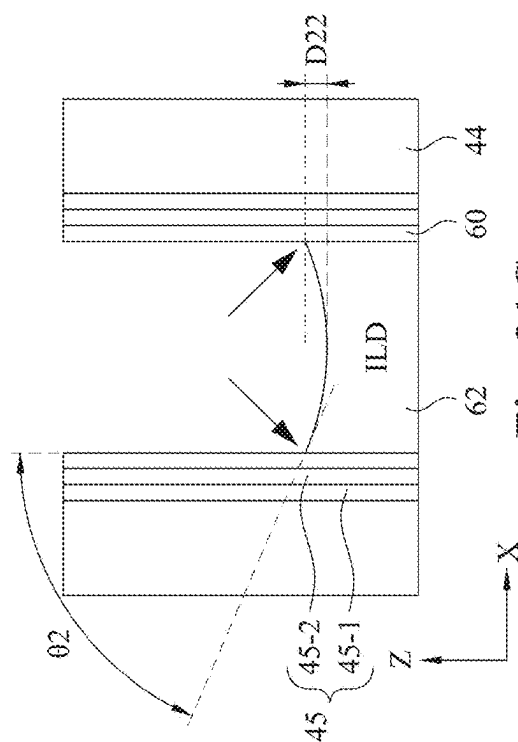


Fig. 21C

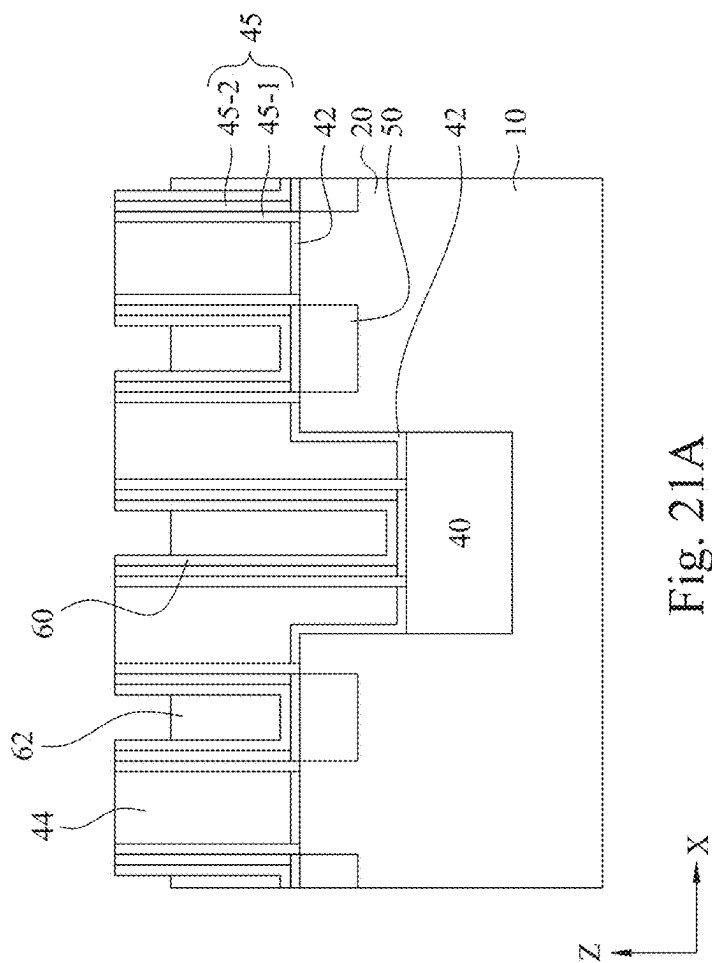
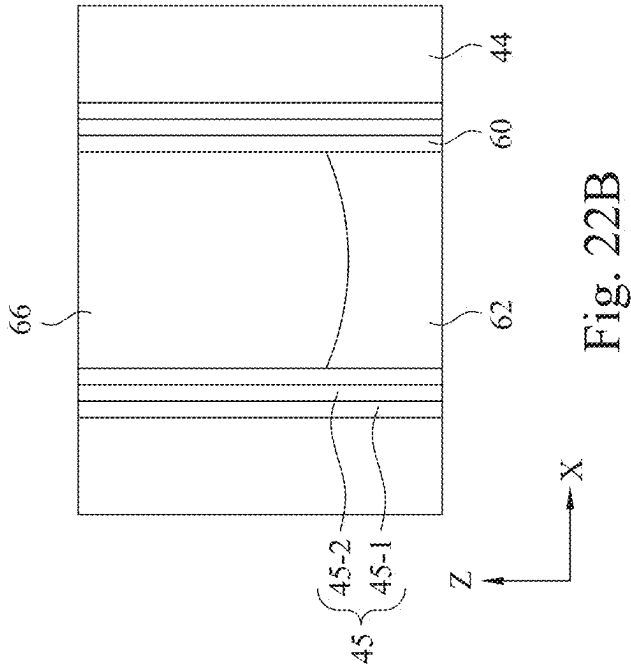
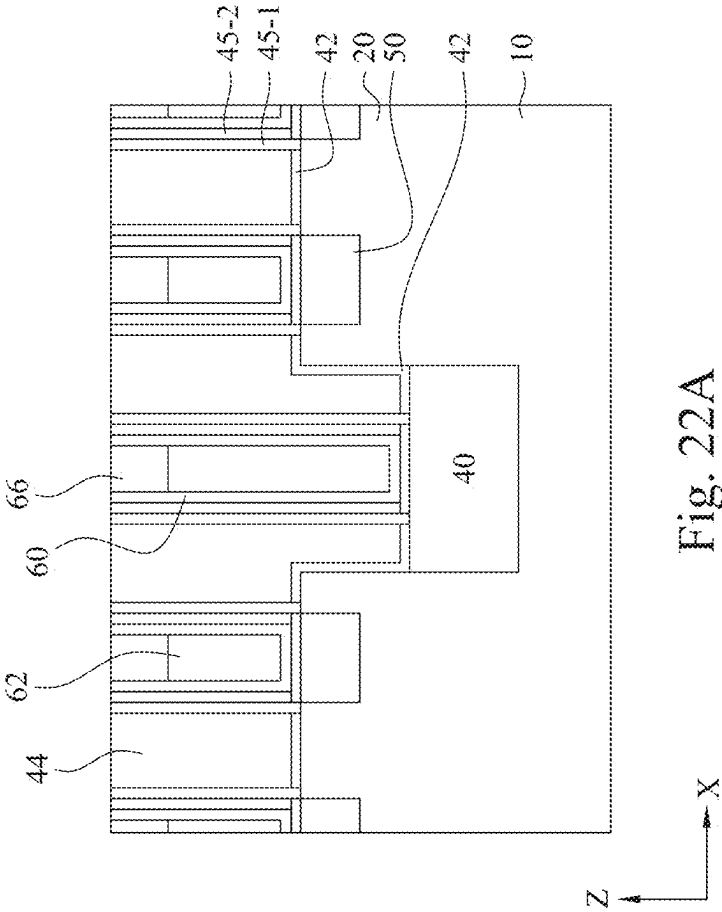
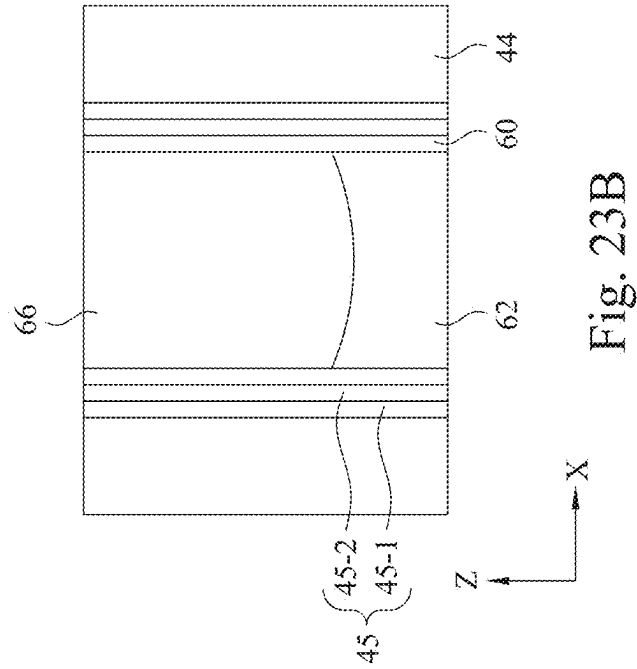
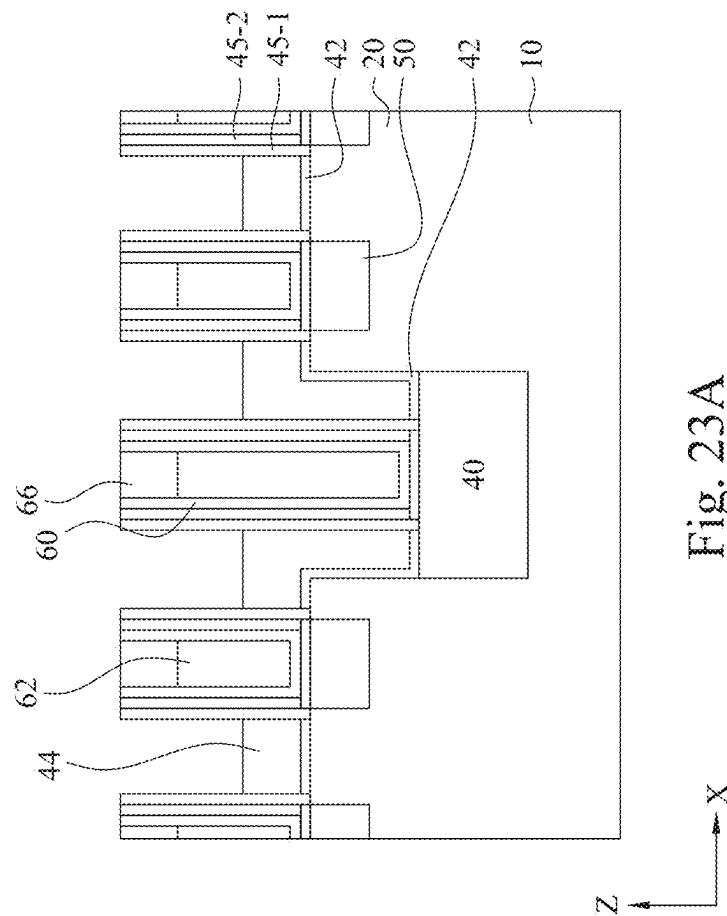


Fig. 21A





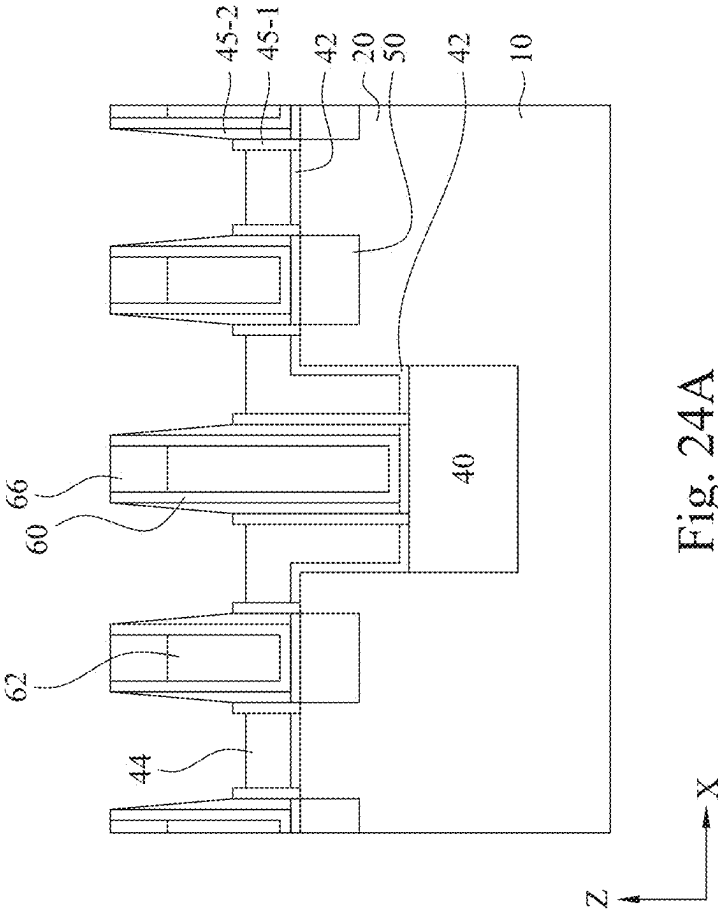


Fig. 24A

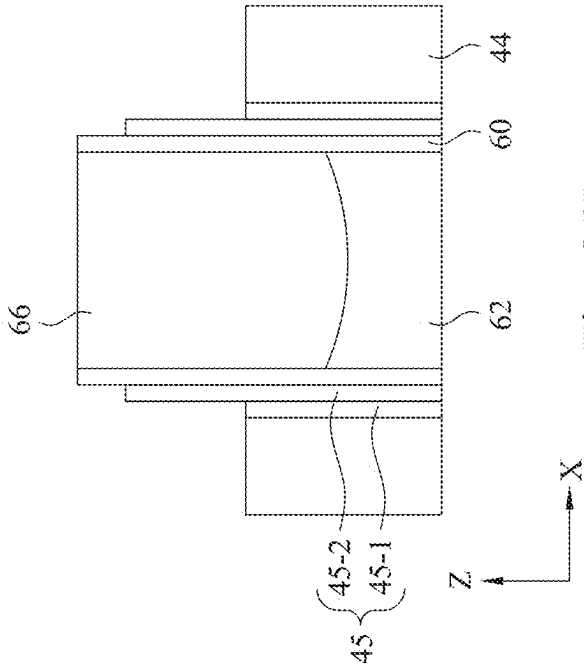
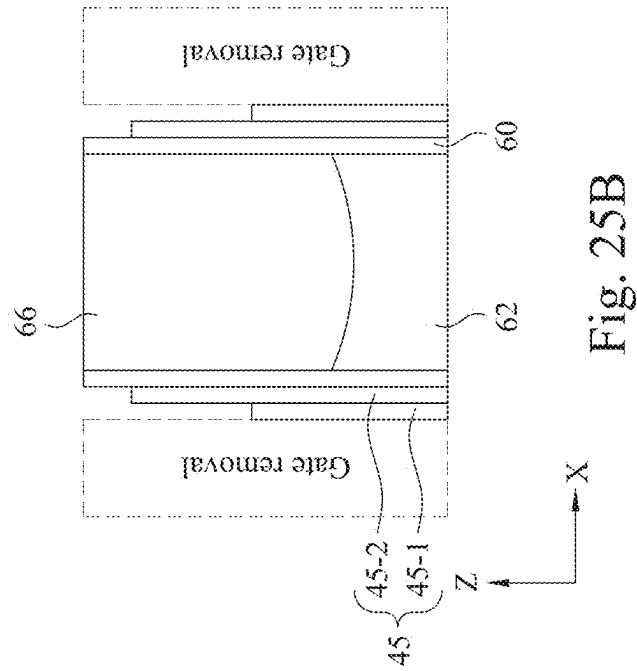
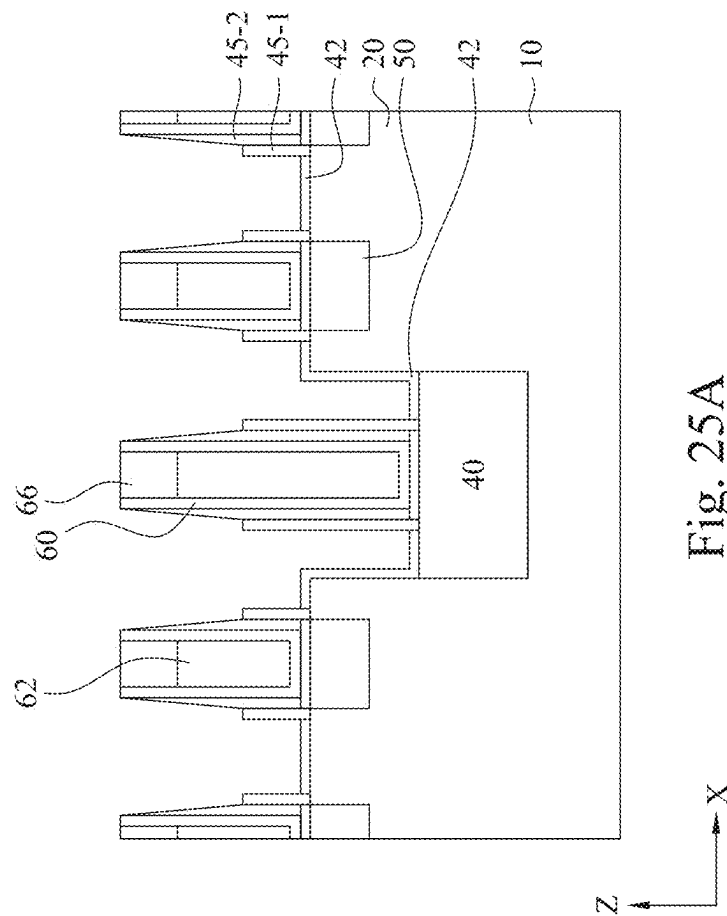


Fig. 24B



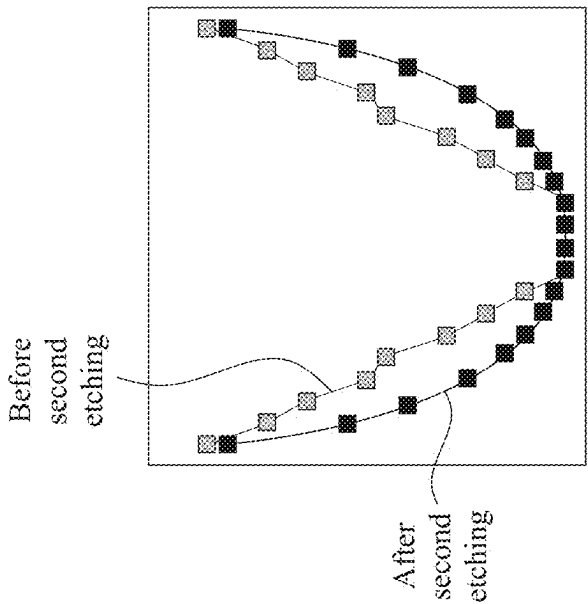


Fig. 26B

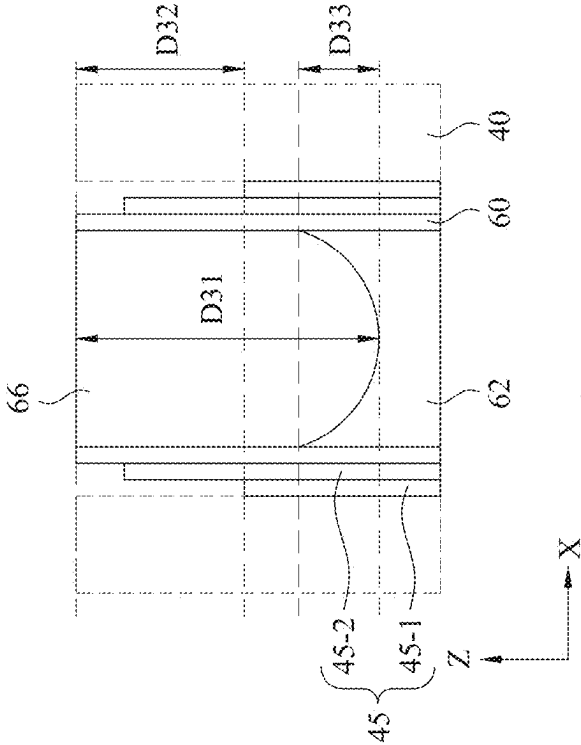


Fig. 26A

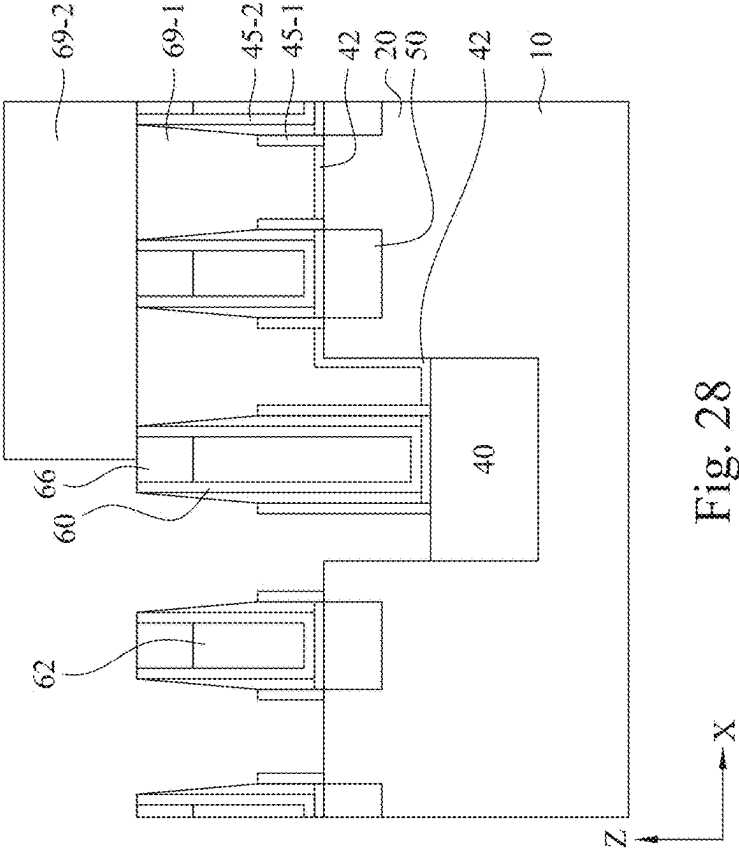


Fig. 28

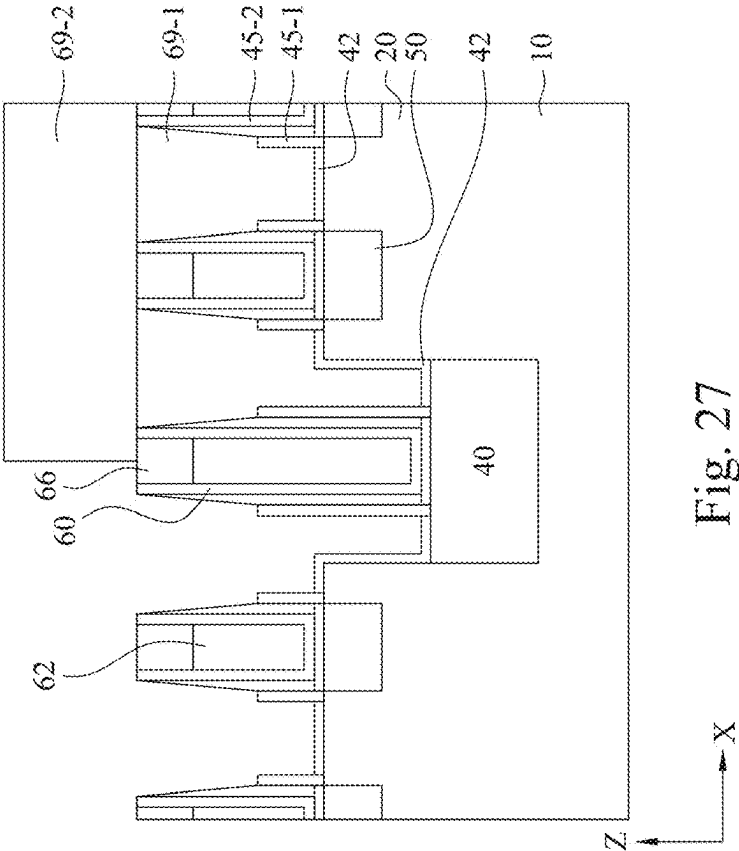
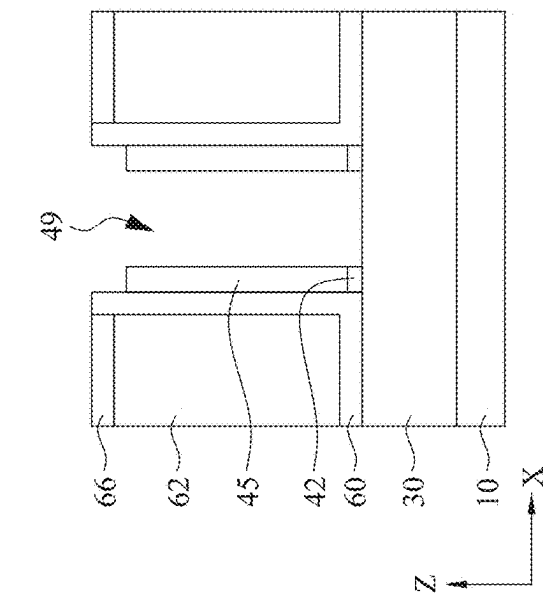
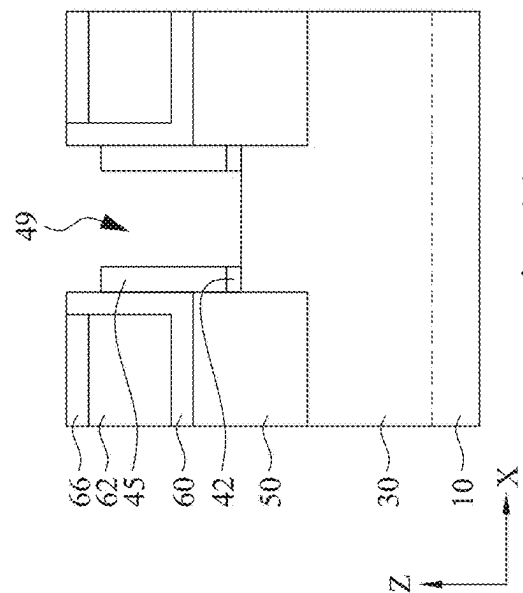
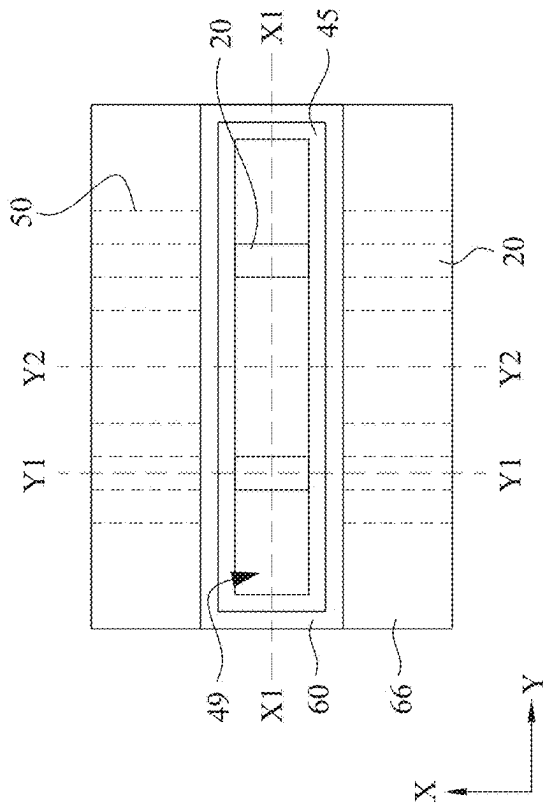
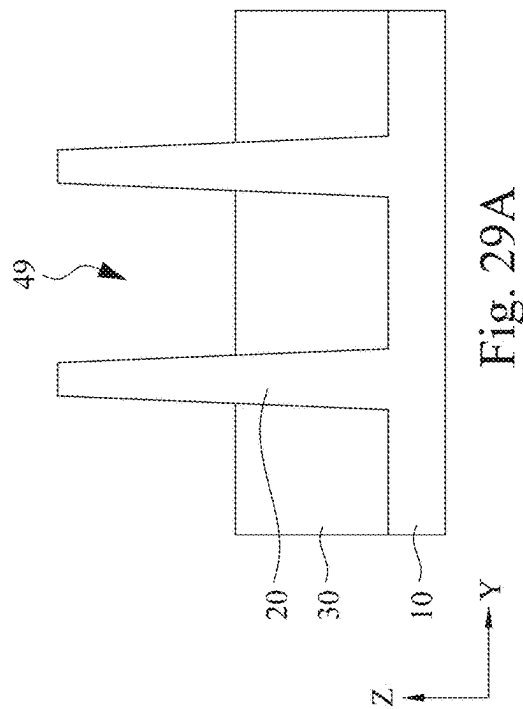


Fig. 27



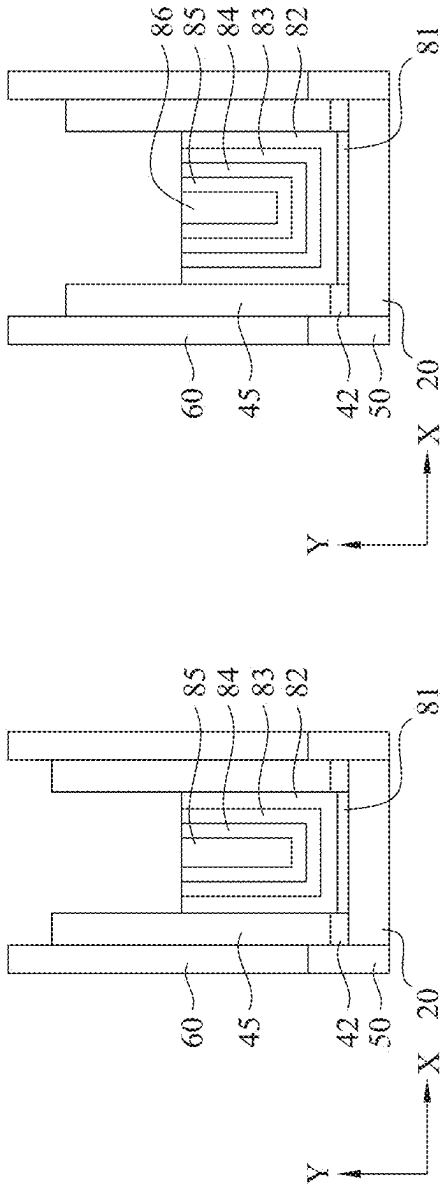


Fig. 31B

Fig. 31A

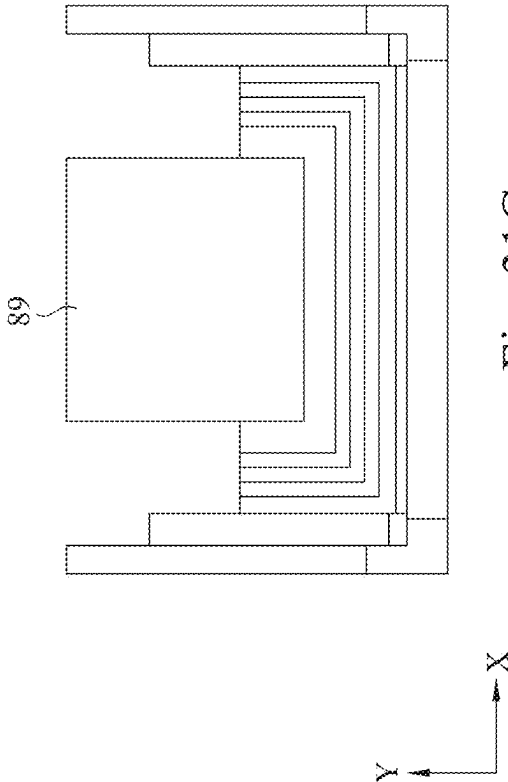
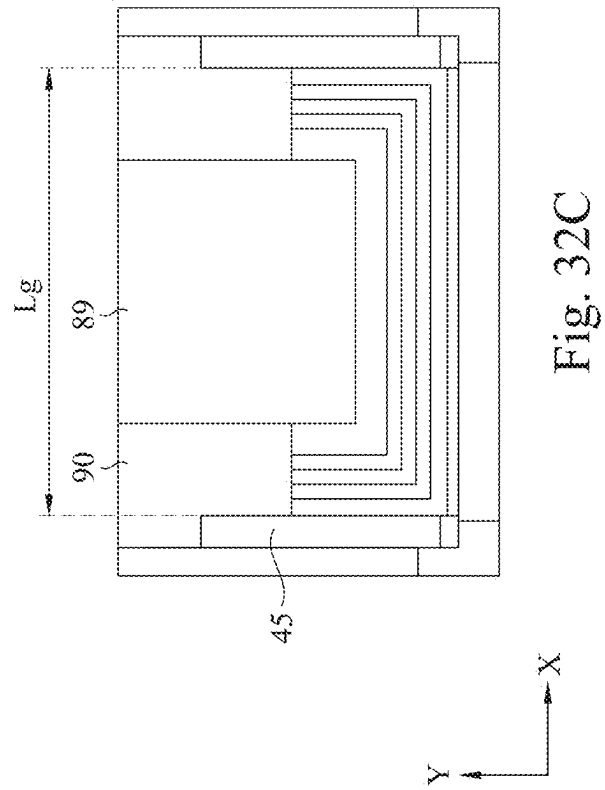
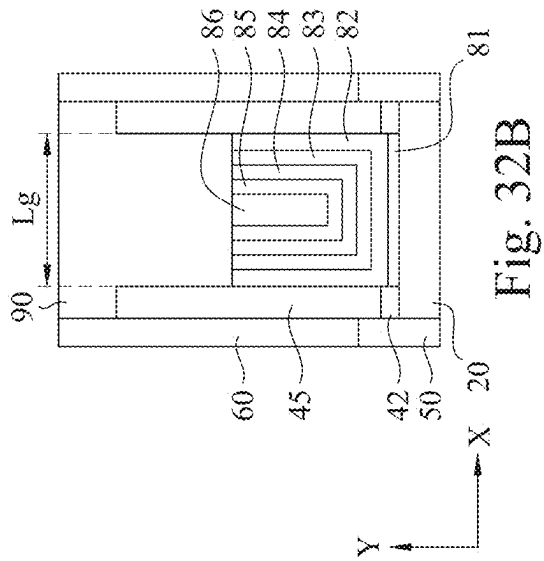
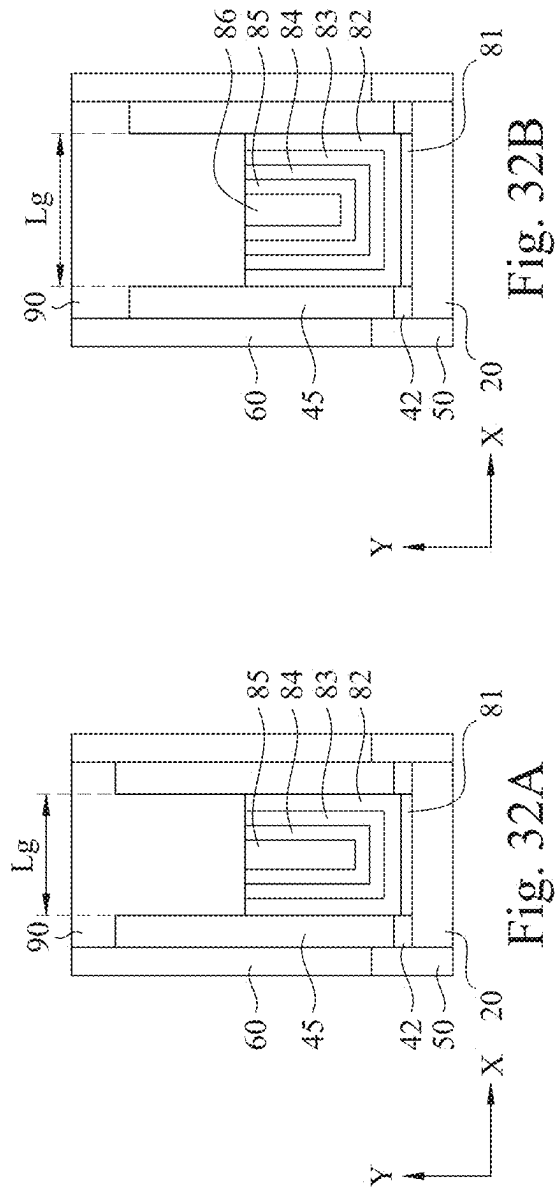


Fig. 31C



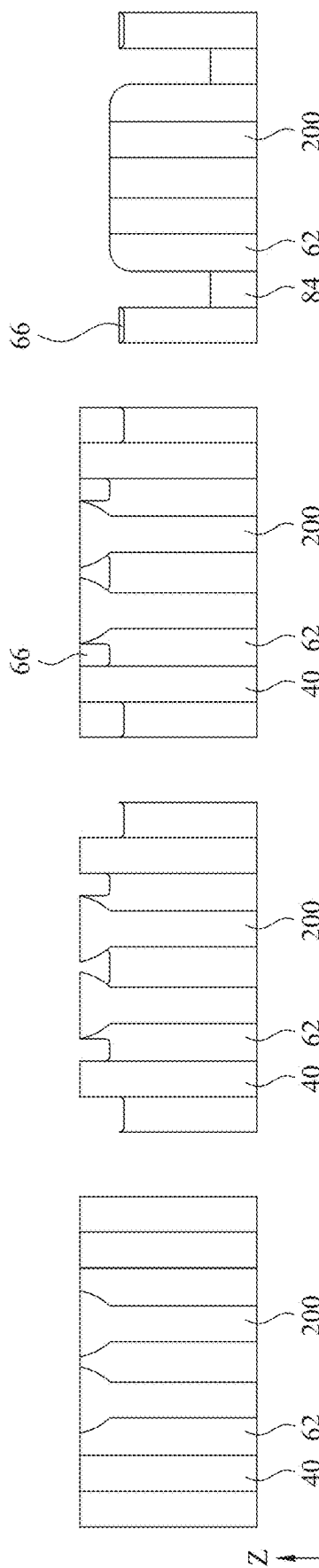
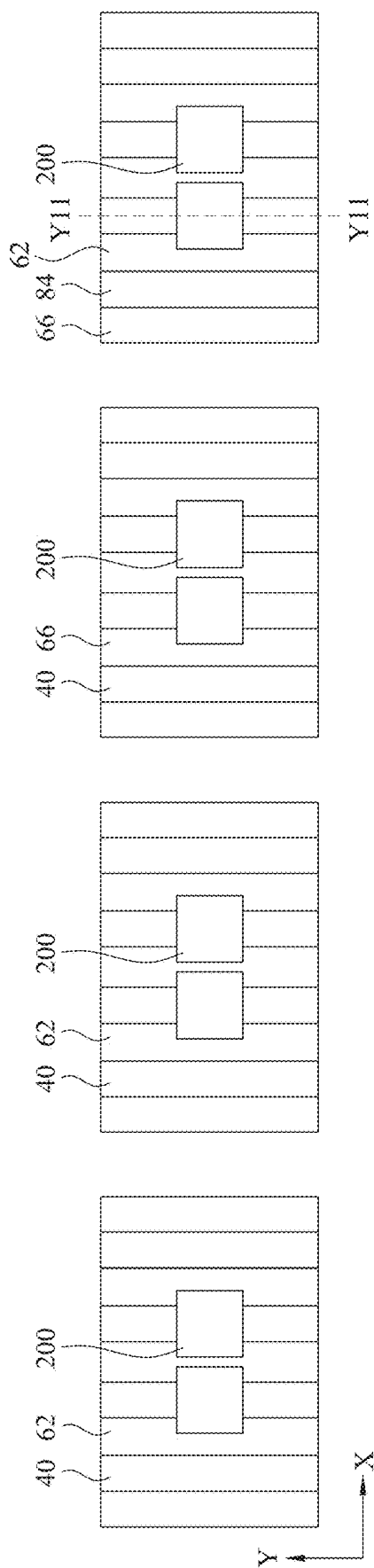


Fig. 33A

Fig. 33B

Fig. 33C

Fig. 33D

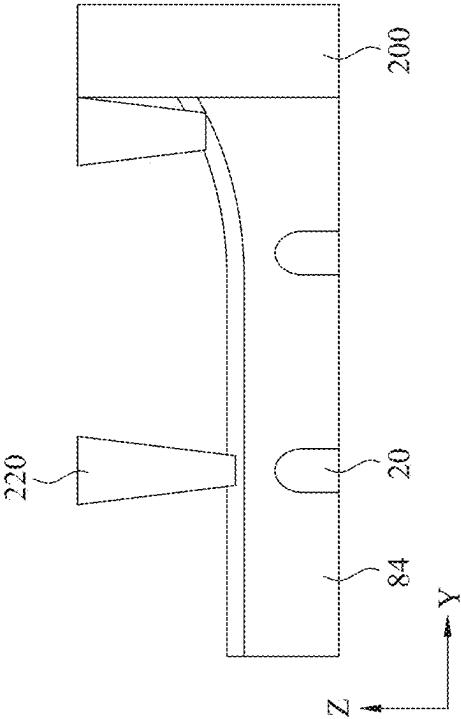


Fig. 35

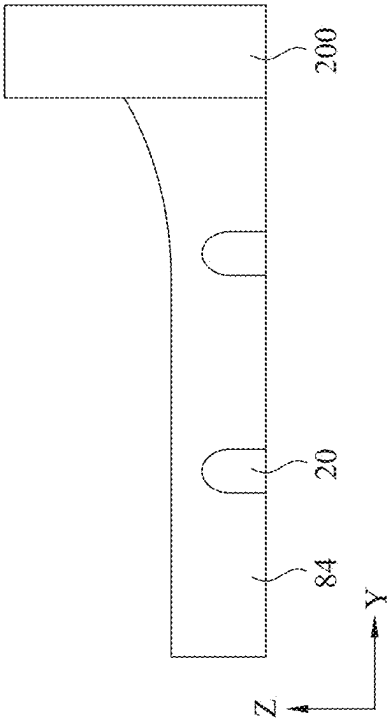


Fig. 34

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METHOD OF MANUFACTURING A SEMICONDUCTOR DEVICE AND A SEMICONDUCTOR DEVICE

RELATED APPLICATION

This application is a divisional application of U.S. application Ser. No. 17/460,097 filed Aug. 27, 2021, the entire disclosure of which is incorporated herein by reference.

BACKGROUND

As the semiconductor industry has progressed into nanometer technology process nodes in pursuit of higher device density, higher performance, and lower costs, challenges from both fabrication and design issues have resulted in the development of three-dimensional designs, such as a multi-gate field effect transistor (FET), including a fin FET (FinFET) and a gate-all-around (GAA) FET. In a FinFET, a gate electrode is adjacent to three side surfaces of a channel region with a gate dielectric layer interposed therebetween. A gate electrode of a FinFET includes one or more layers of metallic material formed by a gate replacement technology.

BRIEF DESCRIPTION OF THE DRAWINGS

The present disclosure is best understood from the following detailed description when read with the accompanying figures. It is emphasized that, in accordance with the standard practice in the industry, various features are not drawn to scale and are used for illustration purposes only. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. 1 shows one of the stages of a sequential process for manufacturing a semiconductor device according to an embodiment of the present disclosure.

FIG. 2 shows one of the stages of a sequential process for manufacturing a semiconductor device according to an embodiment of the present disclosure.

FIG. 3 shows one of the stages of a sequential process for manufacturing a semiconductor device according to an embodiment of the present disclosure.

FIG. 4 shows one of the stages of a sequential process for manufacturing a semiconductor device according to an embodiment of the present disclosure.

FIG. 5 shows one of the stages of a sequential process for manufacturing a semiconductor device according to an embodiment of the present disclosure.

FIG. 6 shows one of the stages of a sequential process for manufacturing a semiconductor device according to an embodiment of the present disclosure.

FIG. 7 shows one of the stages of a sequential process for manufacturing a semiconductor device according to an embodiment of the present disclosure.

FIG. 8 shows one of the stages of a sequential process for manufacturing a semiconductor device according to an embodiment of the present disclosure.

FIG. 9 shows one of the stages of a sequential process for manufacturing a semiconductor device according to an embodiment of the present disclosure.

FIG. 10 shows one of the stages of a sequential process for manufacturing a semiconductor device according to an embodiment of the present disclosure.

FIG. 11 shows one of the stages of a sequential process for manufacturing a semiconductor device according to an embodiment of the present disclosure.

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FIG. 12 shows one of the stages of a sequential process for manufacturing a semiconductor device according to an embodiment of the present disclosure.

FIG. 13 shows one of the stages of a sequential process for manufacturing a semiconductor device according to an embodiment of the present disclosure.

FIG. 14 shows one of the stages of a sequential process for manufacturing a semiconductor device according to an embodiment of the present disclosure.

FIG. 15 shows one of the stages of a sequential process for manufacturing a semiconductor device according to an embodiment of the present disclosure.

FIG. 16 shows one of the stages of a sequential process for manufacturing a semiconductor device according to an embodiment of the present disclosure.

FIGS. 17A, 17B, 17C and 17D show various stages of a sequential process for manufacturing a semiconductor device according to an embodiment of the present disclosure.

FIGS. 18A, 18B, 18C, 18D and 18E show various stages of a sequential process for manufacturing a semiconductor device according to an embodiment of the present disclosure.

FIGS. 19A and 19B show one of the stages of a sequential process for manufacturing a semiconductor device according to an embodiment of the present disclosure.

FIG. 20 shows one of the stages of a sequential process for manufacturing a semiconductor device according to an embodiment of the present disclosure.

FIGS. 21A, 21B and 21C show one of the stages of a sequential process for manufacturing a semiconductor device according to an embodiment of the present disclosure.

FIGS. 22A and 22B show one of the stages of a sequential process for manufacturing a semiconductor device according to an embodiment of the present disclosure.

FIGS. 23A and 23B show one of the stages of a sequential process for manufacturing a semiconductor device according to an embodiment of the present disclosure.

FIGS. 24A and 24B show one of the stages of a sequential process for manufacturing a semiconductor device according to an embodiment of the present disclosure.

FIGS. 25A and 25B show one of the stages of a sequential process for manufacturing a semiconductor device according to an embodiment of the present disclosure.

FIG. 26A shows one of the stages of a sequential process for manufacturing a semiconductor device according to an embodiment of the present disclosure.

FIG. 26B shows surface profile before and after an additional etching operation according to an embodiment of the present disclosure.

FIGS. 27 and 28 show various stages of a sequential process for manufacturing a semiconductor device according to an embodiment of the present disclosure.

FIGS. 29A, 29B, 29C and 29D show one of the various stages of a sequential process for manufacturing a semiconductor device according to an embodiment of the present disclosure.

FIGS. 30A, 30B and 30C show one of the various stages of a sequential process for manufacturing a semiconductor device according to an embodiment of the present disclosure.

FIGS. 31A, 31B and 31C show one of the various stages of a sequential process for manufacturing a semiconductor device according to an embodiment of the present disclosure.

FIGS. 32A, 32B and 32C show one of the various stages of a sequential process for manufacturing a semiconductor device according to an embodiment of the present disclosure.

FIGS. 33A, 33B, 33C and 33D show the various stages of a sequential process for manufacturing a semiconductor device according to an embodiment of the present disclosure.

FIGS. 34 and 35 show the various stages of a sequential process for manufacturing a semiconductor device according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

It is to be understood that the following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific embodiments or examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, dimensions of elements are not limited to the disclosed range or values, but may depend upon process conditions and/or desired properties of the device. Moreover, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed interposing the first and second features, such that the first and second features may not be in direct contact. Various features may be arbitrarily drawn in different scales for simplicity and clarity.

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. In addition, the term “made of” may mean either “comprising” or “consisting of.”

In a gate replacement technology, a sacrificial gate structure including a sacrificial gate electrode (made of, for example, polysilicon) is first formed over a channel region and subsequently is replaced with a metal gate structure. In the gate replacement technology, various planarization operations, such as chemical mechanical polishing processes, are employed to planarize a dielectric layer, a polysilicon layer and/or a metallic layer. Further, in some FinFET devices, after the gate replacement process to form a metal gate structure, an upper portion of the metal gate structure is recessed and a cap insulating layer is formed over the recessed gate structure to secure an isolation between the metal gate electrode and adjacent conductive contacts. In the present disclosure, a method of suppressing a dishing problem in a chemical-mechanical polishing (CMP) operation, and to improve isolation property of the cap insulating layer is provided.

FIGS. 1-16 show a sequential process for manufacturing an FET device according to an embodiment of the present disclosure. It is understood that additional operations can be provided before, during, and after processes shown by FIGS. 1-16, and some of the operations described below can be

replaced or eliminated, for additional embodiments of the method. The order of the operations/processes may be interchangeable.

As shown in FIG. 1, impurity ions (dopants) 12 are implanted into a substrate 10 to form a well region. The ion implantation is performed to prevent a punch-through effect.

In one embodiment, the substrate 10 includes a single crystalline semiconductor layer on at least its surface portion. The substrate 10 may comprise a single crystalline semiconductor material such as, but not limited to Si, Ge, SiGe, GaAs, InSb, GaP, GaSb, InAlAs, InGaAs, GaSbP, GaAsSb and InP. In this embodiment, the substrate 10 is made of Si.

The substrate 10 may include in its surface region, one or more buffer layers (not shown). The buffer layers can serve to gradually change the lattice constant from that of the substrate to that of the source/drain regions. The buffer layers may be formed from epitaxially grown single crystalline semiconductor materials such as, but not limited to Si, Ge, GeSn, SiGe, GaAs, InSb, GaP, GaSb, InAlAs, InGaAs, GaSbP, GaAsSb, GaN, GaP, and InP. In a particular embodiment, the substrate 10 comprises silicon germanium (SiGe) buffer layers epitaxially grown on the silicon substrate 10. The germanium concentration of the SiGe buffer layers may increase from 30 atomic % germanium for the bottom-most buffer layer to 70 atomic % germanium for the top-most buffer layer.

The substrate 10 may include various regions that have been suitably doped with impurities (e.g., p-type or n-type conductivity). The dopants 12 are, for example boron (BF₂) for an n-type Fin FET and phosphorus for a p-type Fin FET.

In FIG. 2, a mask layer 15 is formed over the substrate 10. In some embodiments, the mask layer 15 includes a first mask layer 15A and a second mask layer 15B. In some embodiments, the first mask layer 15A is made of silicon nitride and the second mask layer 15B is made of a silicon oxide. In other embodiments, the first mask layer 15A is made of silicon oxide and the second mask layer 15B is made of silicon nitride (SiN). The first and second mask layers are formed by chemical vapor deposition (CVD), including low pressure CVD (LPCVD) and plasma enhanced CVD (PECVD), physical vapor deposition (PVD), atomic layer deposition (ALD), or other suitable process. The mask layer 15 is patterned into a mask pattern by using patterning operations including photo lithography and etching.

Next, as shown in FIG. 3, the substrate 10 is patterned by using the patterned mask layer 15 into fin structures 20 extending in the X direction. In FIG. 3, two fin structures 20 are arranged in the Y direction. However, the number of the fin structures is not limited to two, and may be as small as one and three or more. In some embodiments, one or more dummy fin structures are formed on both sides of the fin structures 20 to improve pattern fidelity in the patterning operations.

The fin structures 20 may be patterned by any suitable method. For example, the fin structures may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and is patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned pro-

cess. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the fin structures.

After the fin structures are formed, an insulating material layer including one or more layers of insulating material is formed over the substrate so that the fin structures are fully embedded in the insulating layer. The insulating material for the insulating layer may include silicon oxide, silicon nitride, silicon oxynitride (SiON), SiOCN, SiCN, fluorine-doped silicate glass (FSG), or a low-k dielectric material, formed by LPCVD (low pressure chemical vapor deposition), plasma-CVD or flowable CVD. An anneal operation may be performed after the formation of the insulating layer. Then, a planarization operation, such as a chemical mechanical polishing (CMP) method and/or an etch-back method, is performed such that the upper surfaces of the fin structures **20** are exposed from the insulating material layer **30** as shown in FIG. 4.

In some embodiments, one or more liner layers **22** are formed over the structure of FIG. 3 before forming the insulating material layer **30**, as shown FIG. 4. The liner layer **22** includes one or more of silicon nitride, SiON, SiCN, SiOCN, and silicon oxide.

Then, as shown in FIG. 5, the insulating material layer **30** is recessed to form an isolation insulating layer **30** so that the upper portions of the fin structures **20** are exposed. With this operation, the fin structures **20** are electrically separated from each other by the isolation insulating layer **30**, which is also called a shallow trench isolation (STI). The lower portion **11** of the fin structures are embedded in the isolation insulating layer **30**.

After the isolation insulating layer **30** is formed, a sacrificial gate dielectric layer **42** is formed, as shown in FIG. 6. The sacrificial gate dielectric layer **42** includes one or more layers of insulating material, such as a silicon oxide-based material. In one embodiment, silicon oxide formed by CVD is used. The thickness of the sacrificial gate dielectric layer **42** is in a range from about 1 nm to about 5 nm in some embodiments.

FIG. 7 illustrates a structure after a sacrificial gate structure **40** is formed over the exposed fin structures **20**. The sacrificial gate structure includes a sacrificial gate electrode **44** and a sacrificial gate dielectric layer **42**. The sacrificial gate structure **40** is formed over a portion of the fin structures, which is to be a channel region. The sacrificial gate structure **40** is formed by first blanket depositing the sacrificial gate dielectric layer **42** over the fin structures. A sacrificial gate electrode layer is then blanket deposited on the sacrificial gate dielectric layer and over the fin structures, such that the fin structures are fully embedded in the sacrificial gate electrode layer. The sacrificial gate electrode layer includes silicon such as polycrystalline silicon or amorphous silicon. In some embodiments, the sacrificial gate electrode layer is subjected to a planarization operation. The sacrificial gate dielectric layer and the sacrificial gate electrode layer are deposited using CVD, including LPCVD and PECVD, PVD, ALD, or other suitable process. Subsequently, a mask layer is formed over the sacrificial gate electrode layer. The mask layer includes a pad SiN layer **46** and a silicon oxide mask layer **48**.

Next, a patterning operation is performed on the mask layer and sacrificial gate electrode layer is patterned into the sacrificial gate structure **40**, as shown in FIG. 7. The patterning operations of sacrificial gate structure **40** will be explained below in more detail.

The sacrificial gate structure **40** includes the sacrificial gate dielectric layer **42**, the sacrificial gate electrode layer **44** (e.g., poly silicon), the pad SiN layer **46** and the silicon

oxide mask layer **48** in some embodiments. By patterning the sacrificial gate structure **40**, the upper portions of the fin structures **20** are partially exposed on opposite sides of the sacrificial gate structure **40**, thereby defining source/drain (S/D) regions, as shown in FIG. 7. In this disclosure, a source and a drain are interchangeably used and the structures thereof are substantially the same. In FIG. 7, one sacrificial gate structure is formed, but the number of the sacrificial gate structures is not limited to one. Two or more sacrificial gate structures are arranged in the X direction in some embodiments. In certain embodiments, one or more dummy sacrificial gate structures are formed on both sides of the sacrificial gate structures to improve pattern fidelity.

After the sacrificial gate structure **40** is formed, a blanket layer **45** of an insulating material for sidewall spacers **45** is conformally formed by using CVD or other suitable methods, as shown in FIG. 8. The blanket layer **45** is deposited in a conformal manner so that it is formed to have substantially equal thicknesses on vertical surfaces, such as the sidewalls, horizontal surfaces, and the top of the sacrificial gate structure. In some embodiments, the blanket layer **45** is deposited to a thickness in a range from about 2 nm to about 10 nm. In one embodiment, the insulating material of the blanket layer **45** is a silicon nitride-based material, such as SiN, SiON, SiOCN or SiCN and combinations thereof.

Further, as shown in FIG. 9, sidewall spacers **45** are formed on opposite sidewalls of the sacrificial gate structures, and subsequently, the fin structures of the S/D regions are recessed down below the upper surface of the isolation insulating layer **30**. After the blanket layer **45** is formed, anisotropic etching is performed on the blanket layer **45** using, for example, reactive ion etching (RIE). During the anisotropic etching process, most of the insulating material is removed from horizontal surfaces, leaving the dielectric spacer layer on the vertical surfaces such as the sidewalls of the sacrificial gate structures and the sidewalls of the exposed fin structures. The mask layer **48** may be exposed from the sidewall spacers. In some embodiments, isotropic etching may be subsequently performed to remove the insulating material from the upper portions of the S/D region of the exposed fin structures **20**.

Subsequently, the fin structures of the S/D regions are recessed down below the upper surface of the isolation insulating layer **30**, by using dry etching and/or wet etching. As shown in FIG. 9, the sidewall spacers **45** formed on the S/D regions of the exposed fin structures (fin sidewalls) partially remain. In other embodiments, however, the sidewall spacers **45** formed on the S/D regions of the exposed fin structures are fully removed. In case of a GAA FET, inner spacers are formed after the recessing the S/D regions.

Subsequently, as shown in FIG. 10, source/drain (S/D) epitaxial layers **50** are formed. The S/D epitaxial layer **50** includes one or more layers of Si, SiP, SiC and SiCP for an n-channel FET or Si, SiGe, Ge, GeSn and SiGeSn for a p-channel FET. The S/D layers **50** are formed by an epitaxial growth method using CVD, ALD or molecular beam epitaxy (MBE).

As shown in FIG. 10, the S/D epitaxial layers grow from the recessed fin structures respectively. The grown epitaxial layers merge above the isolation insulating layer and form a void **52** in some embodiments.

Subsequently, an insulating liner layer **60**, as an etch stop layer, is formed and then an interlayer dielectric (ILD) layer **65** is formed, as shown in FIG. 11. The insulating liner layer **60** is made of a silicon nitride-based material, such as SiN, and functions as a contact etch stop layer in the subsequent etching operations. The materials for the ILD layer **65**

include compounds comprising Si, O, C and/or H, such as silicon oxide, SiCOH and SiOC. Organic materials, such as polymers, may be used for the ILD layer **65**. After the ILD layer **65** is formed, a planarization operation, such as CMP, is performed, so that the top portion of the sacrificial gate electrode layer **44** is exposed, as shown in FIG. **11**.

Next, as shown in FIG. **12**, the sacrificial gate electrode layer **44** and sacrificial gate dielectric layer **42** are removed, thereby exposing the fin structures in a gate space **49**. The ILD layer **65** protects the S/D structures **50** during the removal of the sacrificial gate structures. The sacrificial gate structures can be removed using plasma dry etching and/or wet etching. When the sacrificial gate electrode layer **44** is polysilicon and the ILD layer **65** is silicon oxide, a wet etchant such as a TMAH solution can be used to selectively remove the sacrificial gate electrode layer **44**. The sacrificial gate dielectric layer **42** is thereafter removed using plasma dry etching and/or wet etching.

After the sacrificial gate structures are removed, a gate dielectric layer **82** is formed around the exposed fin structures **20**, and a gate electrode layer **88** is formed on the gate dielectric layer **82**, as shown in FIG. **13**.

In certain embodiments, the gate dielectric layer **82** includes one or more layers of a dielectric material, such as silicon oxide, silicon nitride, or high-k dielectric material, other suitable dielectric material, and/or combinations thereof. Examples of high-k dielectric material include HfO_2 , HfSiO , HfSiON , HfTaO , HfTiO , HfZrO , zirconium oxide, aluminum oxide, titanium oxide, hafnium dioxide-alumina ($\text{HfO}_2\text{—Al}_2\text{O}_3$) alloy, other suitable high-k dielectric materials, and/or combinations thereof. In some embodiments, the gate dielectric layer **82** includes an interfacial layer formed between the channel layers and the dielectric material.

The gate dielectric layer **82** may be formed by CVD, ALD or any suitable method. In one embodiment, the gate dielectric layer **82** is formed using a highly conformal deposition process such as ALD in order to ensure the formation of a gate dielectric layer having a uniform thickness on the channel regions. The thickness of the gate dielectric layer **82** is in a range from about 1 nm to about 6 nm in some embodiments.

The gate electrode layer **88** is formed on the gate dielectric layer **82**. The gate electrode **88** includes one or more layers of conductive material, such as polysilicon, aluminum, copper, titanium, tantalum, tungsten, cobalt, molybdenum, tantalum nitride, nickel silicide, cobalt silicide, TiN, WN, TiAl, TiAlN, TaCN, TaC, TaSiN, metal alloys, other suitable materials, and/or combinations thereof.

The gate electrode layer **88** may be formed by CVD, ALD, electro-plating, or other suitable method. The gate electrode layer is also deposited over the upper surface of the ILD layer **65**. The gate dielectric layer and the gate electrode layer formed over the ILD layer **65** are then planarized by using, for example, CMP, until the top surface of the ILD layer **65** is revealed.

After the planarization operation, the gate electrode layer **88** is recessed and a cap insulating layer **90** is formed over the recessed gate electrode **88**, as shown in FIG. **13**. In some embodiments, the cap insulating layer **90** includes one or more layers of a silicon nitride-based material, such as SiN. The cap insulating layer **90** can be formed by depositing an insulating material followed by a planarization operation.

In certain embodiments of the present disclosure, one or more work function adjustment layers (not shown) are interposed between the gate dielectric layer **82** and the gate electrode **88**. The work function adjustment layers are made

of a conductive material such as a single layer of TiN, TaN, TaAlC, TiC, TaC, Co, Al, TiAl, HfTi, TiSi, TaSi or TaAlC, or a multilayer of two or more of these materials. For the n-channel FET, one or more of TaN, TaAlC, TiN, TiC, Co, TiAl, HfTi, TiSi and TaSi is used as the work function adjustment layer, and for the p-channel FET, one or more of WN, WCN, W, Ru, Co, TiN or TiSiN is used as the work function adjustment layer. The work function adjustment layer may be formed by ALD, PVD, CVD, e-beam evaporation, or other suitable process. Further, the work function adjustment layer may be formed separately for the n-channel FET and the p-channel FET, which may use different metal layers.

Subsequently, contact holes **110** are formed in the ILD layer **65** by using dry etching, as shown in FIG. **14**. In some embodiments, the upper portion of the S/D epitaxial layer **50** is etched.

A silicide layer **120** is formed over the S/D epitaxial layer **50**, as shown in FIG. **15**. The silicide layer includes one or more of WSi, CoSi, NiSi, TiSi, MoSi and TaSi. Then, a conductive material **130** is formed in the contact holes as shown in FIG. **16**. The conductive material **130** includes one or more of Co, Ni, W, Ti, Ta, Cu, Al, TiN and TaN.

It is understood that the FinFETs undergo further CMOS processes to form various features such as contacts/vias, interconnect metal layers, dielectric layers, passivation layers, etc.

FIGS. **17A-18E** show various views of a detailed sequential process of FIGS. **11-12** according to an embodiment of the present disclosure. It is understood that additional operations can be provided before, during, and after processes shown by FIGS. **17A-18E**, and some of the operations described below can be replaced or eliminated, for additional embodiments of the method. The order of the operations/processes may be interchangeable. Materials, processes, methods, dimensions and/or configurations as explained with the foregoing embodiments may be applied to the following embodiments, and detailed description thereof may be omitted.

After one or more sacrificial gate structures corresponding to FIG. **10** are formed, a first dielectric layer **62** is formed to fully cover the sacrificial gate structures and a second dielectric layer **64** made of a different material than the first dielectric layer **62** is further formed over the first dielectric layer **62**, as shown in FIG. **17A**. In some embodiments, as shown in FIG. **17A**, the sacrificial gate structures include fine patterns corresponding to short channel FETs (e.g., gate length $L_g \leq 20$ nm) and coarse or large patterns corresponding to long channel FETs (e.g., $50 \text{ nm} \leq L_g \leq 500$ nm). Further, in some embodiments, a space between adjacent sacrificial gate structures varies between the same width as the fine patterns to about 2-5 times the width of the fine patterns, to between 50 nm to about 500 nm.

In some embodiments, the first dielectric layer **62** is made of a silicon oxide based material, such as silicon oxide, SiON and SiOC. In some embodiments, the second dielectric layer **64** is made of a silicon nitride based material, such as silicon nitride, SiON and SiCN. In some embodiments, the thickness of the second dielectric layer **64** is smaller than the first dielectric layer **62**. The first and second dielectric layer are formed by LPCVD, plasma-CVD, ALD or any other suitable film formation methods. In some embodiments, no second dielectric layer is formed.

Then, as shown in FIG. **17B**, one or more planarization operations are performed on the first and second dielectric layers to expose the sacrificial gate electrodes **44** (polysilicon layer) of the sacrificial gate structures. In some embodi-

ments, the planarization operation includes a CMP operation. During the planarization operation, the hard mask layers **46** and **48** are also removed.

In some embodiments, the planarization operation includes a first CMP process for mainly etching the second dielectric layer **64** and a subsequent second CMP process for etching the first dielectric layer **62**, which ends when the polysilicon layer of the sacrificial gate electrode is exposed.

Next, as shown in FIG. **17C**, the remaining first dielectric layer **62** is recessed to expose upper portions of the sacrificial gate structures, by one or more etching (e.g., plasma dry etching) operations. In some embodiments, the recessed amount **D11** is about 10% to about 30% of the original depth **D10** of the first dielectric layer **62** after the planarization operation, which is also measured from the top of the sacrificial gate structures. In some embodiments, **D10** is in a range from about 100 nm to about 200 nm. In some embodiments, **D11** is in a range from about 10 nm to about 60 nm and is in a range from about 20 nm to about 35 nm in other embodiments.

Subsequently, as shown in FIG. **17D**, a third dielectric layer **66** is formed to fully cover the sacrificial gate structures and a fourth dielectric layer **68** made of a different material than the third dielectric layer **66** is further formed over the third dielectric layer **66**.

In some embodiments, the third dielectric layer **66** is made of a silicon nitride based material, such as silicon nitride, SiON and SiCN. In some embodiments, the silicon nitride layer is doped with some impurities (diffusion silicon nitride film). In some embodiments, the fourth dielectric layer **68** is made of a silicon oxide based material, such as silicon oxide, SiON, TEOS and SiOC. The third and fourth dielectric layers are formed by LPCVD, plasma-CVD, ALD, flowable CVD or any other suitable film formation methods. In some embodiments, the deposition temperature is in a range from about 400° C. to about 600° C.

In some embodiments, the thickness of the third dielectric layer **66** is smaller than the fourth dielectric layer **68**. In some embodiments, the thickness of the third dielectric layer **66** is in a range from about 50 nm to about 100 nm. In some embodiments, the thickness of the third dielectric layer **66** is 2-3 times the depth of the depth **D11**. When the thickness is smaller than this range, flatness of the dielectric layers after the subsequent planarization operation (CMP) may be insufficient, and when the thickness is larger than this range, some patterns at lower pattern density may suffer from dishing problems and deposition and/or polishing time may increase, which will increase manufacturing cost. The thickness of the fourth dielectric layer **68** is in a range from about 100 nm to about 200 nm in some embodiments to improve flatness after the subsequent planarization (CMP) process.

Then, as shown in FIGS. **18A-18C**, planarization operations are performed on the first and second dielectric layer to expose the sacrificial gate electrodes **44** of the sacrificial gate structures. The planarization operations include a first CMP process, a second CMP process and a third CMP process sequentially performed in this order, in some embodiments. In the first to third CMP operations, different slurries and/or CMP pads are used.

In the first CMP operation, the fourth dielectric layer **68** is mainly etched. The first CMP operation stops at the surface of the third dielectric layer **66** by employing an end point detection technique. In some embodiments, a down force of the CMP head is relatively low, in a range from about more than 0.1 and up to about 2 psi, for all zones to detect the end point and stop on the third dielectric layer **66** to suppress the dishing problem. When the down force is

higher than this range, dishing problems may occur in an oxide rich-area. In some embodiments, the slurry used in the first CMP operation includes an abrasive containing CeO₂, which etches silicon oxide at a high etching rate (e.g., 30-160 nm/min) and does not substantially etch silicon nitride.

In some embodiments, an additional over-polishing (over-etching) is performed for about 10-30 seconds after the endpoint is detected. As shown in FIG. **18A**, residual portions of the fourth dielectric layer remain due to the topography of the upper surface of the third dielectric layer **66** in some embodiments.

The second CMP operation mainly etches the third dielectric layer **66** and stops on the sacrificial gate electrode layer **44** (polysilicon layer) by employing an end point detection technique. In some embodiments, a down force of the CMP head is relatively low in a range from about more than zero and up to about 3 psi. In some embodiments, an additional over-polishing is performed for about 5-15 seconds (or about 3-9% of the main etching time) after the endpoint is detected. When the over-polishing time is too short, the third dielectric layer **66** may remain on the sacrificial gate electrode **44**, and when the over-polishing time is too long, a dishing problem at a large space portion (see, FIG. **18D**) may occur. In some embodiments, in the second CMP operation, the sacrificial gate electrode layer **44** is slightly etched in an amount of 0.5 nm to about 2.5 nm.

The third CMP operation etches both the third dielectric layer **66** and the sacrificial gate electrode layer **44** in some embodiments. The third CMP operation is controlled by time. In some embodiments, the etching time of the third CMP operation is in a range from about 5 sec to about 15 sec. In some embodiments, in the second CMP operation, the sacrificial gate electrode layer **44** is etched in an amount of 0.5 nm to about 5 nm. In some embodiments, the remaining third dielectric layer **66** after the third CMP process is in a range from about 15 nm to about 30 nm. The structure of FIG. **18C** corresponds to the structure of FIG. **11** and the ILD layer **65** includes the first dielectric layer **62** and the remaining third dielectric layer **66**. In some embodiments, the slurry used in the third CMP has a relatively high etching rate of silicon nitride (e.g., 20-50 nm/min) and relatively low polysilicon etching rate. In some embodiments, the thickness **D12** of the remaining third dielectric layer is about 10% to about 30% of the height **D13** of the sacrificial gate structure from the upper surface of the isolation insulating layer, which is about 90 nm to about 180 nm in some embodiments. In some embodiments, **D12** is in a range from about 9 nm to about 54 nm and is in a range from about 20 nm to about 35 nm in other embodiments.

In some embodiments, in a large space portion or coarse pattern portion (distance between adjacent sacrificial gate structures is about 50 nm or more (up to, for example, about 500 nm)) between the sacrificial gate structures, dishing may be observed as shown in FIG. **18D**. In some embodiments, the dishing amount **D20** measured from the top of the adjacent sacrificial gate structures is in a range from about 1 nm to about 5 nm. In some embodiments, the difference between the maximum thickness (observed at, e.g., dense pattern) and the minimum thickness (observed at, e.g., coarse patterns) of the remaining third dielectric layer **66** is in a range from about 2 nm to about 10 nm.

In the first to third CMP processes, CMP pads having high or medium hardness (e.g., hardness >50 (Shore D) are used for high planarization efficiency. When the hardness is less than about 50, the polishing may not yield good planarization efficiency, the dishing effect may occur at a low density

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pattern region, and the third dielectric layer **66** on the first dielectric layer **62** may be removed due to the dishing effect and then would not be able to protect the dielectric layer **62** in subsequent processes. Further, in the first to third CMP processes, since the CMP process mainly etches silicon nitride, a post CMP cleaning process may be performed. The post CMP cleaning process includes a pre-clean process (e.g., platen buffing), an ultrasonic (mega sonic) wafer cleaning process, a brush conditioning process, and an IPA (isopropanol) cleaning process. The brush conditioning process includes a first step and a second step in some embodiments. In some embodiments, the first step includes the use of an acid solution buffing operation with one or more chelating agents to capture metal ions during the polishing, and the second step includes a base solution (e.g., NH_4OH) to remove excessive organic materials.

After the CMP operations, similar to FIG. **12**, the sacrificial gate electrode layer **44** and sacrificial gate dielectric layer **42** are removed, to form gate spaces **49**, as shown in FIG. **18E**.

The foregoing planarization operations of FIGS. **17A-18C** can be applied to planarization operations for other underlying structures, such as fin structures, metal gate electrodes, and contact and wiring structures.

FIGS. **19A-28** show various views of a detailed sequential process of FIGS. **17B-18E** according to an embodiment of the present disclosure. It is understood that additional operations can be provided before, during, and after processes shown by FIGS. **19A-28**, and some of the operations described below can be replaced or eliminated, for additional embodiments of the method. The order of the operations/processes may be interchangeable. Materials, processes, methods, dimensions and/or configurations as explained with the foregoing embodiments may be applied to the following embodiments, and detailed description thereof may be omitted.

FIGS. **19A** and **19B** show a perspective view and a plan view after the sacrificial gate structures **40** (sacrificial gate electrode **44** and sacrificial gate dielectric layer **42**) are removed, respectively. In some embodiments, one or more sacrificial gate structures **40** are formed to fully overlap one or more fin structures **20**, and a sacrificial gate structure **40** is formed on an edge of the fin structure to cover the edge of the fin structure.

FIG. **20** shows a cross section view showing sacrificial gate structures fully overlapping the fin structure and a sacrificial gate structure over the edge of the fin structure **20**, after the insulating liner layer **60** (an etch stop layer) and the ILD layer **62** are formed, similar to FIG. **17B**. In some embodiments, the gate sidewall spacers **45** include first sidewall spacers **45-1** formed on side faces of the sacrificial gate electrode layer **44** and second sidewall spacers **45-2** made of a different material than the first sidewall spacers **45-1** formed over the first sidewall spacers **45-1**, as shown in FIG. **20**. In some embodiments, the second sidewall spacers **45-2** are continuously formed between the adjacent sacrificial gate structures as shown in FIG. **20**, and in other embodiments, the second sidewall spacers **45-2** have no bottom portion between the adjacent sacrificial gate structures. In some embodiments, the sacrificial gate dielectric layer **42** extends below the first and/or second sidewall spacers.

In some embodiments, the first sidewall spacers **45-1** are made of an insulating material having a higher nitrogen concentration (and a lower oxide concentration) than an insulating material of the second sidewall spacers **45-2**. In some embodiments, the first and second sidewall spacers

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include one or more of SiOCN , SiCN , SiN , SiON , SiOC or any other suitable insulating materials. In some embodiments, the thickness of the first sidewall spacer and the second sidewall spacer is in a range from about 1 nm to about 10 nm, respectively, depending on the design and/or process requirements.

FIG. **21A** shows a cross sectional view corresponding to FIG. **17C** after the remaining first dielectric (ILD) layer **62** is recessed to expose upper portions of the sacrificial gate structures. In some embodiments, the recess etching of the ILD layer **62** includes two or more etching processes, e.g., a first etching process and a second etching process subsequent to the first etching process, with different etching conditions from each other. In some embodiments, the first etching process is a chemical dry etching process without using plasma. The source gas for the chemical dry etching includes HF and NH_3 . In some embodiments, an etching selectivity (etching rate ratio) of silicon oxide (ILD **62**) to silicon nitride is about 40:1 to about 60:1. FIG. **21B** shows an enlarged cross-sectional view of the upper surface of the etched ILD layer **62** after the first etching process. As shown in FIG. **21B**, the center portion of the ILD layer **62** is etched more than the edge portions and the upper surface of the etched ILD layer **62** has a large peak-to-valley distance **D21**. In some embodiments, the peak-to-valley distance **D21** (a distance between the highest point and the lowest point) of the ILD **62** is in a range from about 2 nm to about 15 nm. In some embodiments, an angle $\theta 1$ between the upper surface of the ILD layer **62** and a side face of the second sidewall spacer **45-2** at the highest contacting point of the ILD layer **62** to the second sidewall spacer **45-2** is in a range from about 10 degrees to about 50 degrees, and is in a range from about 20 degrees to about 30 degrees in other embodiments.

In some embodiments, the second etching process is a plasma dry etching process. The source gas for the chemical dry etching includes NF_3 and H_2 . In some embodiments, the plasma process uses a remote plasma source. In some embodiments, an etching selectivity (etching rate ratio) of silicon oxide (ILD **62**) to silicon nitride is about 3:1 to about 5:1. FIG. **21C** shows an enlarged cross-sectional view of the upper surface of the etched ILD layer **62** after the second etching process. As shown in FIG. **21C**, the edge portions are etched to reduce the peak-to-valley distance. In some embodiments, the peak-to-valley distance **D22** after the second etching process is smaller than **D21** and is in a range from about 1 nm to about 10 nm. In some embodiments, an angle $\theta 2$ between the upper surface of the ILD layer **62** and a side face of the second sidewall spacer **45-2** at the highest contacting point of the ILD layer **62** to the second sidewall spacer **45-2** is in a range from about 60 degrees to about 95 degrees, and is in a range from about 70 degrees to about 90 degrees.

FIGS. **22A** and **22B** show cross sectional views corresponding to FIGS. **18B-18D** after the third dielectric layer **66** is formed.

FIGS. **23A** and **23B** show cross sectional views after the sacrificial gate electrode layer **44** is partially recessed. In some embodiments, a plasma dry etching process is used to remove the sacrificial gate electrode layer **44**. In some embodiments, about 40% to about 70% in height above the upper surface of the fin structure **20** of the sacrificial gate electrode layer **44** is recessed (etched).

FIGS. **24A** and **24B** show cross sectional views after the gate sidewall spacers **45** are partially recessed. In some embodiments, the first sidewall spacers **45-1** are recessed to the substantially same level as the recessed sacrificial gate

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electrode layer **44** ($\pm$ about 5 nm). In some embodiments, the second sidewall spacers **45-2** are also partially recessed. The recessed amount of the second sidewall spacers **45-2** is smaller than the recessed amount of the first sidewall spacers **45-1** and is about 20% to about 40% of the recessed amount of the first sidewall spacers.

FIGS. **25A** and **25B** show cross sectional views corresponding to FIG. **18E** after the sacrificial gate electrode layer **44** is fully removed. In some embodiments, a plasma dry etching process and/or a wet etching process are used to remove the sacrificial gate electrode layer **44**.

FIG. **26A** shows an enlarged view of the third insulating layer **66** and the gate sidewall spacers. In some embodiments, the dimension **D33** is smaller than dimensions **D31-D32**, where the dimension **D31** is a largest thickness of the third insulating layer **66** above the ILD layer **62**, the dimension **D32** is a recess amount of the second sidewall spacers **45-2** and the dimension **D33** is a peak-to-valley distance of the upper surface of the ILD layer **62**. When the dimensions do not satisfy $D33 < D31 - D32$, the metal gate electrode **88** (shown in FIGS. **13-16**) may have a horn profile which may degrade device performance and/or process margins.

FIG. **26B** shows a profile comparison of the upper surface of the ILD layer **62** before and after the second etching process. The vertical axis is normalized. As shown in FIG. **26B**, the steep V-shape profile is relaxed to the U-shape profile by the second etching process as set forth above.

After the sacrificial gate electrode **44** is fully removed, an I/O region having a larger gate length is covered by a cover layer **69-1** and a mask layer **69-2** are formed in the I/O region, as shown in FIG. **27**. In some embodiments, the cover layer **69-1** is made of a dielectric material and the mask layer **69-2** is a photo resist pattern. Then, as shown in FIG. **28**, the sacrificial gate dielectric layer **42** in the core region having a smaller gate length is removed. Subsequently, the mask layer **69-2** and the cover layer **69-1** are removed, and metal gate structures as explained with respect to FIG. **13** are formed in the core region and the I/O region, respectively.

FIGS. **29A-32C** show various views of a detailed sequential process of FIGS. **12-13** according to an embodiment of the present disclosure. It is understood that additional operations can be provided before, during, and after processes shown by FIGS. **29A-32C**, and some of the operations described below can be replaced or eliminated, for additional embodiments of the method. The order of the operations/processes may be interchangeable. Materials, processes, methods, dimensions and/or configurations as explained with the foregoing embodiments may be applied to the following embodiments, and detailed description thereof may be omitted.

FIGS. **29A-29D** show various views after the sacrificial gate structure (sacrificial gate electrode **44** and sacrificial gate dielectric layer **42**) are removed, thereby forming a gate space **49**, as described with reference to FIG. **12**. FIG. **29A** is a cross sectional view along X1-X1 of FIG. **29D** (a plan or projected view), FIG. **29B** is a cross sectional view along Y1-Y1 of FIG. **29D** and FIG. **29C** is a cross sectional view along Y2-Y2 of FIG. **29D**. In some embodiments, an insulating liner layer **60** functioning as an etching stop layer is formed before the ILD layer **65** is formed. In some embodiments, the insulating liner layer **60** includes silicon nitride.

In some embodiments, an upper portion of the gate sidewall spacer **45** is recessed as shown in FIGS. **29B** and **29C**. In some embodiments, the gate sidewall spacers **45** are recessed during the removal of the sacrificial gate dielectric layer, and in other embodiments, one or more dry and/or wet

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etching operations are performed to recess the gate sidewall spacers **45**. In some embodiments, after the gate sidewall spacers **45** are recessed, the uppermost surface is made of only a silicon nitride based material (e.g., silicon nitride) (layers **60** and **66**).

FIGS. **30A**, **30B** and **30C** are enlarged views of the gate spaces after the metal gate structures are formed. FIG. **30A** corresponds to a short channel FET including gate electrodes of static random access memory (SRAM) cells (e.g., $4 \text{ nm} \leq L_g \leq 10 \text{ nm}$), FIG. **30B** corresponds to a middle channel FET including gate electrodes of logic circuits (e.g., $12 \text{ nm} \leq L_g \leq 20 \text{ nm}$), and FIG. **30C** corresponds to a long channel FET ($50 \text{ nm} \leq L_g \leq 500 \text{ nm}$).

As shown in FIGS. **30A-30C**, an interfacial layer **81** is formed on the channel regions of the fin structures **20** and a gate dielectric layer **82** is formed over the interfacial layer and inner wall of the gate sidewall spacers **45**. In some embodiments, the gate dielectric layer **82** is formed over the upper surfaces of the etching stop layer **60** and the additional dielectric layer **66**. The gate dielectric layer **82** is formed by an ALD process in some embodiments to conformally form a layer over a high aspect ratio structure. In some embodiments, the aspect ratio (height/bottom diameter or area) of the gate space **49** is in a range from about 7 to about 25.

Then, a barrier layer **83** is formed over the gate dielectric layer **82**. In some embodiments, the barrier layer **83** includes one or more layers of Ta, TaN, Ti, TiN or TiSiN. In some embodiments, the thickness of the barrier layer is in a range from about 1 nm to about 3 nm. In some embodiments, the barrier layer **83** is not formed. In some embodiments, the thickness of the barrier layer **83** at the bottom is thicker than the thickness at the sides. In some embodiments, the thickness of the barrier layer **83** at the bottom is about 0.5 times to three times the thickness at the sides.

Further, one or more first work function adjustment material (WFM) layers are formed over the barrier layer **83**. In some embodiments, the first WFM layer **84** is a p-type WFM material, such as WN, WCN, W, Ru, Co, TiN or TiSiN. In some embodiments, the thickness of the first WFM layer is in a range from about 0.5 nm to about 10 nm and is in a range from about 1 nm to about 2 nm in other embodiments. In some embodiments, the thickness of the first WFM layer **84** at the bottom is about 0.8 times to twice the thickness at the side. When the first WFM layer is made of TiN, the TiN layer is formed from source gases including TiCl_4 and NH_3 . In some embodiments, the TiN layer contains Cl as an impurity. In some embodiments, the Ti concentration in the TiN layer is in a range from about 10 atomic % to about 80 atomic %. When the Ti concentration is too small, the resistance of the TiN layer increases, and when the Ti concentration is too high, Ti diffusion may cause various problems (e.g., punch-through).

Further, one or more second WFM layers **85** are formed over the first WFM layer **84**. In some embodiments, the second WFM layer **85** is an n-type WFM material, such as TiAl, TiSiAl, TiAlC, TaAl or TaAlC. In some embodiments, the thickness of the second WFM layer is in a range from about 0.5 nm to about 6 nm and is in a range from about 2 nm to about 5 nm in other embodiments. In some embodiments, the thickness of the second WFM layer **85** at the bottom is the same as or up to three times the thickness at the side. After the WFM layers are formed, a body metal layer **86** is formed over the WFM layers. In some embodiments, a glue layer (not shown) is formed over the WFM layers before the body metal layer is formed. In some embodiments, the glue layer includes one or more of Ta, WCN, TaN, Ti, TiN or TiSiN. The body metal layer **86**

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includes W, Ta, Sn, Nb, Ru, Co or Mo. In certain embodiments, W is used. In some embodiments, the body metal layer **86** is formed by an ALD process using metal halide (chloride) gases (e.g., WCl_5 , TaCl_5 , SnCl_4 , NbCl_5 or MoCl_4). In some embodiments, the body metal layer **86** includes fluorine-free metal, for example, fluorine-free W formed by WCl_5 as a source gas. In some embodiments, in an n-type FET, the first WFM layer (p-type material layer) is not formed.

In some embodiments, as shown in FIG. **30A**, the second WFM layer **85** fully fills the gate space and no body metal layer (e.g., W layer) is formed in the gate space. Further, in some embodiments, the conductive layers are conformally formed in the gate space and thus do not fully fill the gate space for a long channel FET as shown in FIG. **30C**. In such a case, one or more insulating layer **89** is formed to fill the remaining space of the gate space, as shown in FIG. **30C**. The insulating layer **89** is formed by one or more deposition and CMP operations. In some embodiments, the insulating layer **89** includes silicon nitride.

Then, as shown in FIGS. **31A-31C**, an upper portion of the conductive layer formed in the gate space is recessed by one or more etching operations. In some embodiments, in the etching operations, an upper part of the sidewall spacers **45** and/or an upper portion of the gate dielectric layer **82** are also etched. In some embodiments, the tops of the WFM layers are lower than the top of the gate dielectric layer **82**. In other embodiments, the top of the body metal layer **86** is higher than the top of either or both of the WFM layers. In the long channel FET, the insulating layer **89** is not substantially etched, as shown in FIG. **31C**.

Further, as shown in FIGS. **32A-32C**, a gate cap insulating layer **90** is formed in over the recessed conductive layers and the recessed gate dielectric layer **82**. In some embodiments, the gate cap insulating layer **90** includes silicon nitride, SiON and/or SiOCN or any other suitable material. In a long channel FET as shown in FIG. **32C**, the gate cap insulating layer **90** fills the space between the gate sidewall spacer **45** and the insulating layer **89**. In some embodiments, the conductive layers **83-86** of the gate electrode and the gate dielectric layer **82** have a U-shape cross section.

FIGS. **33A-33D** show various views of a detailed sequential process of FIGS. **11-13** for forming recesses for the gate cap insulating layers **90** according to an embodiment of the present disclosure. It is understood that additional operations can be provided before, during, and after processes shown by FIGS. **33A-33D**, and some of the operations described below can be replaced or eliminated, for additional embodiments of the method. The order of the operations/processes may be interchangeable. Materials, processes, methods, dimensions and/or configurations as explained with the foregoing embodiments may be applied to the following embodiments, and detailed description thereof may be omitted.

In some embodiments, after a plurality of sacrificial gate structures **40** are formed, part of the sacrificial gate electrodes are cut or etched to form one or more spaces, and the spaces are filled by a dielectric material **200** as shown in FIG. **33A**, which shows a plan view (top) and a cross sectional view (bottom). In some embodiments, part of the fin structures below the sacrificial gate structures is also etched. The dielectric material **200** includes one or more of silicon nitride, SiON or any other suitable dielectric material. As shown in FIG. **33A**, an upper portion of the filled dielectric material **200** has a funnel shape in some embodiments.

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FIG. **33B** corresponds to FIGS. **17C**, **21A** and **21C**, and FIG. **33C** corresponds to FIGS. **18B-18D** and **22A-22B**. In some embodiments, due to the funnel shape of the dielectric material **200**, the recessed amount of the ILD layer **62** adjacent to the dielectric material **200** is smaller than the recessed amount between the sacrificial gate structures **40**.

FIG. **33D** corresponds to FIGS. **31A-31C**. After the metal gate structure **84** is recessed, the funnel shape of the dielectric material **200** and the third insulating layer **66** adjacent to the dielectric material **200** are removed in some embodiments.

FIG. **34** is a cross sectional view after the metal gate electrode layer **84** is recessed, and FIG. **35** shows after contacts **220** to the gate electrode layer **84** is formed, both of which correspond to line Y11-Y11 of FIG. **33D**. As shown in FIG. **34**, the recessed metal gate electrode layer **84** has a slight "horn" shape at or near the interface to the dielectric material **200**. In the present disclosure, however, since the ILD layer **62** is recessed by using two step etching processed to suppress the peak-to-valley distance, the horn height is reduced or suppressed so as not to cause any defects or performance degradation.

The various embodiments or examples described herein offer several advantages over the existing art. In the embodiments of the present disclosure, since the ILD layer between adjacent gate structures has a flatter upper surface by using two step etching operations, it is possible to improve various device performances and yield of the manufactured semiconductor devices.

It will be understood that not all advantages have been necessarily discussed herein, no particular advantage is required for all embodiments or examples, and other embodiments or examples may offer different advantages.

In accordance with one aspect of the present disclosure, in a method of manufacturing a semiconductor device, a sacrificial gate structure is formed over a substrate. The sacrificial gate structure includes a sacrificial gate electrode. A first dielectric layer is formed over the sacrificial gate structure. A second dielectric layer is formed over the first dielectric layer. The second and first dielectric layers are planarized. The second and first dielectric layers are recessed such that an upper portion of the sacrificial gate structure is exposed while a lower portion of the sacrificial gate structure being embedded in the first dielectric layer. A third dielectric layer is formed over the exposed sacrificial gate structure and over the first dielectric layer. A fourth dielectric layer is formed over the third dielectric layer. The fourth and third dielectric layers are planarized such that the sacrificial gate electrode is exposed and part of the third dielectric layer remains on the recessed first dielectric layer. In the recessing the first dielectric layer, a first etching operation and a second etching operation using a different etching as from the first etching operation are performed. In one or more of the foregoing and following embodiments, the first etching operation is a chemical dry etching process not using plasma and the second etching operation is a plasma dry etching process. In one or more of the foregoing and following embodiments, an etching gas of the chemical dry etching process includes HF and NH_3 . In one or more of the foregoing and following embodiments, an etching gas of the plasma dry etching process includes NF_3 and H_2 . In one or more of the foregoing and following embodiments, the first dielectric layer includes a silicon oxide based material, and the second dielectric layer includes a silicon nitride based material different from the first dielectric layer. In one or more of the foregoing and following embodiments, the third dielectric layer includes a silicon nitride based mate-

rial, and the fourth dielectric layer includes a silicon oxide based material different from the third dielectric layer. In one or more of the foregoing and following embodiments, the planarizing the fourth and third dielectric layers includes: a first chemical mechanical polishing (CMP) process for etching the fourth dielectric layer; a second CMP process for etching the third dielectric layer, which ends when the sacrificial gate electrode is exposed; and a third CMP process for recessing the third dielectric layer and the sacrificial gate electrode. In one or more of the foregoing and following embodiments, the first CMP process comprises a first end point detection and a first over-polishing after detection of the first end point, the second CMP process comprises a second end point detection and a second over-polishing after detection of the second end point, and the third CMP process is time-controlled without using an end point detection. In one or more of the foregoing and following embodiments, the second over-polishing is performed for 5-15 seconds.

In accordance with another aspect of the present disclosure, in a method of manufacturing a semiconductor device, sacrificial gate structures and a first dielectric layer are formed over a substrate. Each of the sacrificial gate structures includes a sacrificial gate electrode, the sacrificial gate structures are embedded in the first dielectric layer and an upper surface of each of the sacrificial gate structures is exposed from the first dielectric layer. The first dielectric layer is partially removed. A second dielectric layer is formed over the partially removed first dielectric layer. The sacrificial gate electrode is removed from each of the sacrificial gate structures, thereby forming gate spaces. Gate electrode structures are formed in the gate spaces. After the first gate dielectric layer is partially removed, a peak-to-valley distance of the partially removed first dielectric layer between two adjacent sacrificial gate structures is in a range from 1 nm to 10 nm. In one or more of the foregoing and following embodiments, in the partially removing the first dielectric layer, a first etching operation and a second etching operation using a different etching as from the first etching operation are performed. In one or more of the foregoing and following embodiments, after the first etching operation, a peak-to-valley distance of the partially removed first dielectric layer between two adjacent sacrificial gate structures is in a range from 2 nm to 15 nm, which is greater than a peak-to-valley distance after the second etching operation. In one or more of the foregoing and following embodiments, an angle of an upper surface of the dielectric layer after the second etching operation at an interface to one of the two adjacent sacrificial gate structures is 60 degrees to 95 degrees with respect to the one of the two adjacent sacrificial gate structures. In one or more of the foregoing and following embodiments, an angle of an upper surface of the dielectric layer after the first etching operation and before the second etching operation at an interface to one of the two adjacent sacrificial gate structures is 10 degrees to 50 degrees with respect to the one of the two adjacent sacrificial gate structures. In one or more of the foregoing and following embodiments, each of the sacrificial gate structures includes a sacrificial gate dielectric layer, first gate sidewall spacers disposed on opposing side faces of the sacrificial gate electrode and second gate sidewall spacers disposed on the first gate sidewall spacers. In one or more of the foregoing and following embodiments, the first gate sidewall spacers have a higher nitrogen concentration than the second gate sidewall spacers. In one or more of the foregoing

ing and following embodiments, an etch stop layer is further formed between the first dielectric layer and the second gate sidewall spacers.

In accordance with another aspect of the present disclosure, in a method of manufacturing a semiconductor device, sacrificial gate structures and a first dielectric layer are formed over a substrate. Each of the sacrificial gate structures includes a sacrificial gate electrode, a sacrificial gate dielectric layer, first gate sidewall spacers disposed on opposing side faces of the sacrificial gate electrode and second gate sidewall spacers disposed on the first gate sidewall spacers, the sacrificial gate structures are embedded in the first dielectric layer and an upper surface of each of the sacrificial gate structures is exposed from the first dielectric layer. The first dielectric layer is partially removed. A second dielectric layer is formed over the partially removed first dielectric layer. The sacrificial gate electrode is partially removed from each of the sacrificial gate structures. The second gate sidewall spacers are partially removed. A remaining sacrificial gate electrode is fully removed. The sacrificial gate dielectric layer is removed, thereby forming a gate space. A gate electrode structure is formed in the gate space. In one or more of the foregoing and following embodiments, in the partially removing the first dielectric layer, a first etching operation and a second etching operation using a different etching as from the first etching operation are performed. In one or more of the foregoing and following embodiments, the first etching operation is a chemical dry etching process using HF and NH₃ and the second etching operation is a plasma dry etching process using NF₃ and H₂.

The foregoing outlines features of several embodiments or examples so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments or examples introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A method of manufacturing a semiconductor device, comprising:
 - forming sacrificial gate structures and a first dielectric layer over a substrate, wherein each of the sacrificial gate structures includes a sacrificial gate electrode, the sacrificial gate structures are embedded in the first dielectric layer and an upper surface of each of the sacrificial gate structures is exposed from the first dielectric layer;
 - partially removing the first dielectric layer;
 - forming a second dielectric layer over the partially removed first dielectric layer;
 - removing the sacrificial gate electrode from each of the sacrificial gate structures, thereby forming gate spaces; and
 - forming gate electrode structures in the gate spaces, wherein after the first dielectric layer is partially removed, a peak-to-valley distance of the partially removed first dielectric layer between two adjacent sacrificial gate structures is in a range from 1 nm to 10 nm.

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2. The method of claim 1, wherein the partially removing the first dielectric layer comprises a first etching operation and a second etching operation using a different etchant than the first etching operation.

3. The method of claim 2, wherein after the first etching operation, a peak-to-valley distance of the partially removed first dielectric layer between two adjacent sacrificial gate structures is in a range from 2 nm to 15 nm, which is greater than a peak-to-valley distance after the second etching operation.

4. The method of claim 3, wherein an angle of an upper surface of the first dielectric layer after the second etching operation at an interface to one of the two adjacent sacrificial gate structures is 60 degrees to 95 degrees with respect to the one of the two adjacent sacrificial gate structures.

5. The method of claim 3, wherein an angle of an upper surface of the first dielectric layer after the first etching operation and before the second etching operation at an interface to one of the two adjacent sacrificial gate structures is 10 degrees to 50 degrees with respect to the one of the two adjacent sacrificial gate structures.

6. The method of claim 1, wherein each of the sacrificial gate structures includes a sacrificial gate dielectric layer, first gate sidewall spacers disposed on opposing side faces of the sacrificial gate electrode and second gate sidewall spacers disposed on the first gate sidewall spacers.

7. The method of claim 6, wherein the first gate sidewall spacers have a higher nitrogen concentration than the second gate sidewall spacers.

8. The method of claim 6, wherein an etch stop layer is further formed between the first dielectric layer and the second gate sidewall spacers.

9. A method of manufacturing a semiconductor device, comprising:

forming sacrificial gate structures and a first dielectric layer over a substrate, wherein each of the sacrificial gate structures includes a sacrificial gate electrode, a sacrificial gate dielectric layer, first gate sidewall spacers disposed on opposing side faces of the sacrificial gate electrode and second gate sidewall spacers disposed on the first gate sidewall spacers, the sacrificial gate structures are embedded in the first dielectric layer and an upper surface of each of the sacrificial gate structures is exposed from the first dielectric layer; partially removing the first dielectric layer; forming a second dielectric layer over the partially removed first dielectric layer; partially removing the sacrificial gate electrode from each of the sacrificial gate structures; partially removing the second gate sidewall spacers; fully removing a remaining sacrificial gate electrode; removing the sacrificial gate dielectric layer, thereby forming a gate space; and forming a gate electrode structure in the gate space.

10. The method of claim 9, wherein the partially removing the first dielectric layer comprises a first etching operation and a second etching operation using a different etchant than the first etching operation.

11. The method of claim 10, wherein the first etching operation is a chemical dry etching process using HF and NH₃ and the second etching operation is a plasma dry etching process using NF₃ and H₂.

12. The method of claim 10, wherein an angle of an upper surface of the first dielectric layer after the second etching operation at an interface to one of two adjacent sacrificial gate structures is 60 degrees to 95 degrees with respect to the one of two adjacent sacrificial gate structures.

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13. The method of claim 10, wherein an angle of an upper surface of the first dielectric layer after the first etching operation and before the second etching operation at an interface to one of two adjacent sacrificial gate structures is 10 degrees to 50 degrees with respect to the one of the two adjacent sacrificial gate structures.

14. The method of claim 9, wherein the first gate sidewall spacers have a higher nitrogen concentration than the second gate sidewall spacers.

15. A method of manufacturing a semiconductor device, comprising:

forming a first dielectric layer over sacrificial gate structures formed over a substrate, wherein each of the sacrificial gate structures includes a sacrificial gate electrode;

forming a second dielectric layer over the first dielectric layer;

removing the second dielectric layer;

planarizing the first dielectric layer to expose an upper surface of the sacrificial gate electrode;

recessing the first dielectric layer to expose an upper portion of each of the sacrificial gate structures while a lower portion of each of the sacrificial gate structures is embedded in the first dielectric layer;

forming a third dielectric layer over the exposed upper portion of each of the sacrificial gate structures and the recessed first dielectric layer;

forming a fourth dielectric layer over the third dielectric layer;

removing the fourth dielectric layer; and

planarizing the third dielectric layer to expose the upper surface of the sacrificial gate electrode while part of the third dielectric layer remains on the recessed first dielectric layer,

wherein the recessing the first dielectric layer comprises a first etching operation and a second etching operation using a different etchant than the first etching operation, and

wherein after the first dielectric layer is recessed, a first peak-to-valley distance of the recessed first dielectric layer between two adjacent sacrificial gate structures is in a range from 1 nm to 10 nm.

16. The method of claim 15, wherein each of the sacrificial gate structures further includes a sacrificial gate dielectric.

17. The method of claim 15, wherein the first peak-to-valley distance of the recessed first dielectric layer between the two adjacent sacrificial gate structures after the first etching operation is greater than a second peak-to-valley distance of the recessed first dielectric layer between the two adjacent sacrificial gate structures after the second etching operation.

18. The method of claim 15, wherein an angle of an upper surface of the first dielectric layer after the second etching operation at an interface to one of the two adjacent sacrificial gate structures is 60 degrees to 95 degrees with respect to the one of the two adjacent sacrificial gate structures.

19. The method of claim 15, wherein an angle of an upper surface of the first dielectric layer after the first etching operation and before the second etching operation at an interface to one of the two adjacent sacrificial gate structures is 10 degrees to 50 degrees with respect to the one of the two adjacent sacrificial gate structures.

20. The method of claim 15, wherein the first etching operation is a chemical dry etching process using HF and

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NH₃ and the second etching operation is a plasma dry etching process using NF₃ and H₂.

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